

3

Energy Efficiency in Building and Transportation

In this chapter, we will learn about

- the physical quantities describing the energy efficiency of buildings
- the factors affecting the energy efficiency of buildings
- the main structures of electric vehicles and hybrid vehicles
- the advantages of public transportation systems

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3.1 Energy efficiency in buildings

In Hong Kong, about 30% of electricity is used for air conditioning (Fig 3.1a). As air conditioning produces a cooler interior, heat is naturally transferred from the outside to the interior. If we can reduce this heat transfer, less electricity will be needed for air conditioning and the energy efficiency of the building can be increased.

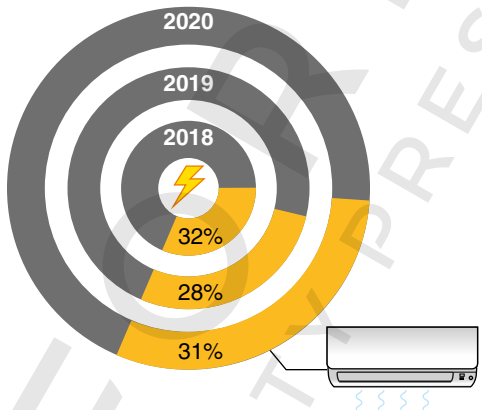


Fig 3.1a *Electricity consumption for air conditioning in Hong Kong. (Data obtained from EMSD website in Oct 2022)*

1 Heat transfer in buildings

Heat flows into or out of a building through the building envelope, which refers to the outermost layer of a building and consists of windows, walls, roofs, etc. There are two ways for heat to transfer through the building envelope (Fig 3.1b).

Windows are assumed to be closed in this chapter.

Cross-link

You learned the conduction and radiation of heat in Book 1 Ch 2.

1 **Conduction** through walls, roof and glass windows

2 **Radiation** through glass windows

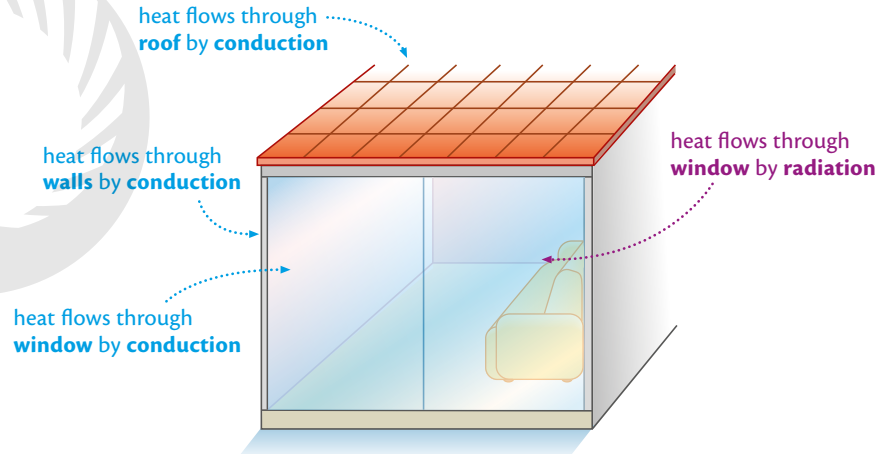


Fig 3.1b *Heat transfer through the building envelope in different ways.*

Simulation

→ This simulation shows the heat transfer in an airconditioned flat. It is available on OUP web in PC version (.exe) only.

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DSE goal

Interpret $\frac{Q}{t} = \kappa \frac{A(T_H - T_C)}{d}$ as the rate of energy transfer by conduction and discuss the heat loss in conduction.

DSE exam

13(2)Q3.5, 18(2)Q3.5

Simulation

→ This simulation demonstrates how different factors affect the rate of conduction. It is available on OUP web in PC version (.exe) only.

2 Heat transfer by conduction

a Rate of heat transfer by conduction

Consider a building with air conditioning. Its inside is colder than the outside. Heat is transferred from the outside to the inside through the building envelope. Consider a wall of the building facing the outside (Fig 3.1c).

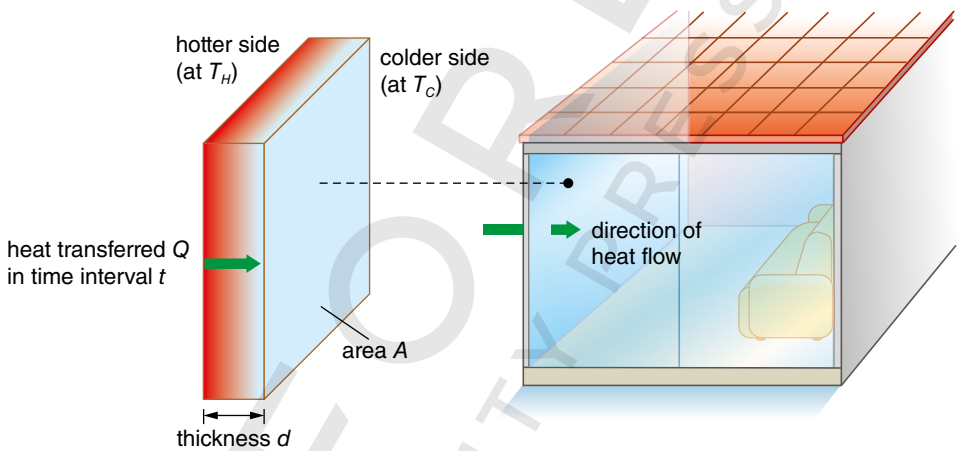


Fig 3.1c Conduction through a wall.

The **rate of heat transfer by conduction** is low when

1. the temperature difference $(T_H - T_C)$ between the hotter and the colder sides of the wall is small;
2. the thickness d of the wall is large;
3. the area A of the wall is small;
4. the wall is made from a good insulator.

Assuming that the wall is made from only a single material, we can express the relation between the **rate of heat transfer by conduction** $\left(\frac{Q}{t}\right)$ and the above factors as follows:

rate of heat transfer (W) = $\frac{\text{heat transfer (J)}}{\text{time interval (s)}} = \frac{Q}{t} = \kappa \frac{A(T_H - T_C)}{d}$

where A = area of surface (m^2); $T_H - T_C$ = temperature difference between two surfaces (K); d = thickness of wall (m); κ = thermal conductivity of material (W m^{-1} K^{-1})

▶ where κ is a constant called **thermal conductivity**, measured in W m^{-1} K^{-1}. It is a physical property of the **material** and describes the ability of a material to conduct heat. A material with smaller thermal conductivity conducts heat more slowly and, therefore, is a better thermal insulator.

κ is pronounced as kappa.

$$\kappa = \frac{Qd}{A(T_H - T_C)t}$$

$$\text{Unit of } \kappa = \frac{\text{J} \times \text{m}}{\text{m}^2 \times \text{K} \times \text{s}} = \frac{\text{J}}{\text{s}} \times \frac{1}{\text{m}} \times \frac{1}{\text{K}} = \text{W m}^{-1} \text{K}^{-1}$$

thermal conductivity 導熱率

Experiment 3a

Estimating the thermal conductivity of glass (p.190)

Video

→ This video demonstrates Experiment 3a.

Figure 3.1d shows the thermal conductivity of some common materials.



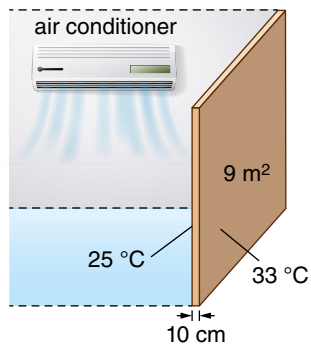
The thermal conductivities of concrete, glass and brick are about the same. ▶

✎ The data can be obtained from the *Code of Practice for Overall Thermal Transfer Value in Buildings*. The document can be found at:

Fig 3.1d *Thermal conductivity of some common materials. (Data obtained from Building Department website in Oct 2022)*

Example 1 Rate of heat transfer by conduction

An air conditioned room has a concrete wall facing outside. The wall is 10 cm thick and has an area of 9 m^2. The temperatures of the inner and the outer surfaces of the wall are 25 °C and 33 °C respectively. The thermal conductivity of the concrete is 1.10 W m^{-1} K^{-1}. Find the rate of heat transfer by conduction through the wall.



Solution

Rate of heat transfer

$$= \kappa \frac{A(T_H - T_C)}{d} = 1.10 \times \frac{9 \times (33 - 25)}{0.1} = 792 \text{ W}$$

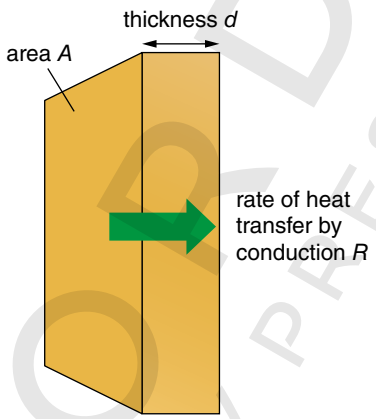
Google Forms Microsoft Forms

Quick check 1

→ E-version in OUP Exercise Platform is also available.

1 *True or false:* A material with a higher thermal conductivity is a better conductor of heat. (T/F)

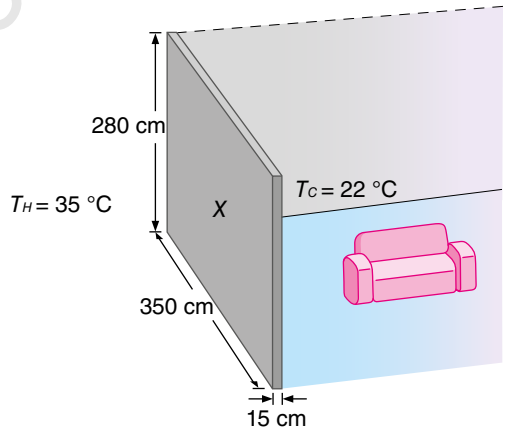
2 The three walls listed below are made of a single material. They transfer heat from one side to the other by conduction. Complete the table.



	Wall	A / m^2	d / m	ΔT / K	κ / W m^{-1} K^{-1}	R / W
(a)	X	4	0.02	5	1.05	
(b)	Y		0.2	7	0.95	960
(c)	Z	20		8	0.44	1500

Learn by practice 1

1 Consider the concrete wall X of a room shown in the figure. The thermal conductivity of the concrete is 1.10 W m^{-1} K^{-1}.



(a) What is the rate of heat flow through X by conduction?

(b) How much energy enters the room through X in 8 hours? Express the answer in kW h.

DSE goal

Define U-value of building materials and solve related problems.

DSE exam

16(2)Q3.6, 17(2)Q.3(b), 19(2)Q3.5, 20(2)Q3.5, Q3.6, 21(2)Q3.5, 22(2)Q3.6, 23(2)Q3.4, etc.

b Thermal transmittance (U-value)

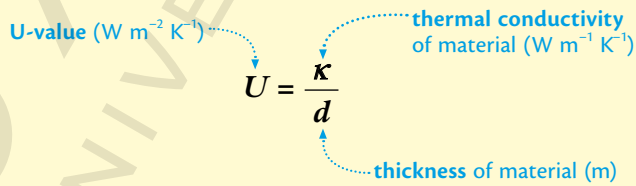
Some building envelopes have complex structures or made up of multiple layers of different materials (Fig 3.1e). In this case, we would consider the **thermal transmittance** (also known as **U-value**) when calculating the heat transfer. The symbol of U-value is U.

(i) (ii)



Fig 3.1e *(i) Bricks with complex structure and (ii) a wall with multiple layers.*

Consider a building envelope made up of a single material only. Its thickness is d and the thermal conductivity of the material is κ. Its U-value is defined as



The SI unit of U-value is W m^{-2} K^{-1}.

Note that κ is not affected by the thickness of the material. It is solely determined by the material. However, ***U* is determined by both the material and thickness.**

The rate of heat transfer by conduction can be related to U-value as follows:

$$\frac{Q}{t} = \kappa \frac{A(T_H - T_C)}{d} = UA(T_H - T_C)$$

The U-value can also be defined as the overall rate of heat transfer through a unit area of a building envelope, divided by the temperature difference across the envelope.

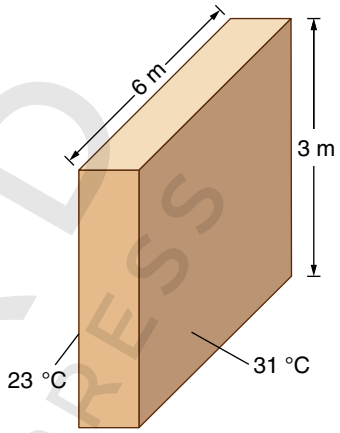
U-value allows us to directly compare the rate of heat conducted away in **all kinds of building envelopes**. The lower the U-value, the lower the rate of heat transfer. Therefore, a building material with a lower U-value is a better thermal insulator.

thermal transmittance (U-value) 熱傳送係數（U 值）

Example 2 Rate of heat transfer and U-value

(a) The size of an insulating wall is 6 m × 3 m. The temperatures of the two surfaces of the wall are 31 °C and 23 °C respectively. What is the rate of heat transfer through the wall? The U-value of the wall is 0.61 W m^{-2} K^{-1}.

(b) If the wall is replaced by another wall of the same size, but with a higher U-value, how will the rate of heat transfer through the wall change?



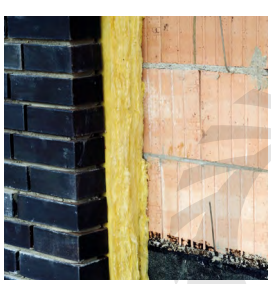
Solution

(a) Area of the wall = 6 × 3
= 18 m^2

Rate of heat transfer = $UA(T_H - T_C)$
= 0.61 × 18 × (31 − 23)
= 87.8 W

(b) Increase

Table 3.1a shows the typical U-values of some common components of a building envelope.



Double-glazed window and triple-glazed window will be discussed on p.100.

	Component of building envelope	U-value / W m^{-2} K^{-1}
Wall	Concrete wall (20 cm thick)	5.5
	Brick wall (20 cm thick)	2.6
	Brick wall (30 cm thick)	2.0
	Insulated wall (mineral wool between bricks)	0.3
Window	Single-glazed window	5.6
	Double-glazed window (12 mm gap with air in between)	2.8
	Triple-glazed window (12 mm gap with air in between)	1.9

Table 3.1a *Typical U-values of some components of a building envelope.*

Calculating the U-value is complicated and the equation $U = \frac{\kappa}{d}$ is only an approximation. Therefore, the values listed in Table 3.1a are not the same as those found using the values in Fig 3.1d and this equation.

For a building envelope made of multiple layers of different materials, the overall U-value can be calculated as shown in Example 3.

U-values for building components normally include all effects of heat transfer through conduction, convection and radiation, during indirect radiation exposure (i.e. the material is not being shone on directly by sunlight or other light sources). Therefore, a window with low-e coating has a lower U-value than a window without the coating.
However, for simplicity, we usually ignore the effect of convection and radiation in discussing U-value in this book.

Example 3 U-value of a composite wall

A wall is composed of a concrete layer and a wood fibre layer. They have different thicknesses and there is no air gap between them. The two sides of the wall are maintained at 32 °C and 25 °C respectively.

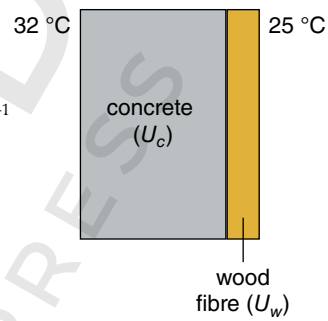
Given:

U-value of the concrete layer = 5.5 W m^{-2} K^{-1}

U-value of the wood fibre layer = 0.95 W m^{-2} K^{-1}

(a) Find the U-value of the wall.

(b) Find the temperature at the junction of the two layers.



Solution

(a) The temperature change across the whole wall equals the sum of the temperature changes across the two layers.

$$\Delta T = \Delta T_c + \Delta T_w \quad \text{..... (1)}$$

When the system is steady, the rate of heat flowing through the concrete layer must be equal to that through the wood fibre layer since there is no energy accumulated at any point. ▶

The rate of heat transfer through each layer and through the whole wall should be the same. Let the rate be r.

For the whole wall: $r = UA\Delta T \Rightarrow \Delta T = \dfrac{r}{UA}$

For the concrete layer: $r = U_c A\Delta T_c \Rightarrow \Delta T_c = \dfrac{r}{U_c A}$

For the wood fibre layer: $r = U_w A\Delta T_w \Rightarrow \Delta T_w = \dfrac{r}{U_w A}$

Therefore, (1) becomes:

$$\frac{r}{UA} = \frac{r}{U_c A} + \frac{r}{U_w A}$$

$$\frac{1}{U} = \frac{1}{U_c} + \frac{1}{U_w}$$

$$= \frac{1}{5.5} + \frac{1}{0.95}$$

$$U = 0.810 \text{ W } m^{-2} \text{ } K^{-1}$$

(b) Let T_j be the temperature at the junction of the two layers.

Rate of heat transfer in concrete = rate of heat transfer in wood fibre

$$U_c A_c \Delta T_c = U_w A_w \Delta T_w$$

$$(5.5)(A)(32 - T_j) = (0.95)(A)(T_j - 25)$$

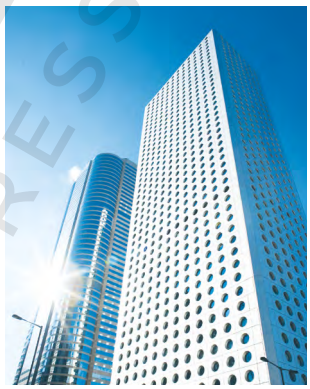
$$T_j = 31.0 \text{ °C}$$

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Different parts of a building envelope are made of different materials. By knowing the U-values of these parts, the total rate of heat transfer by conduction through the envelope can be found.

Example 4 Total rate of heat transfer by conduction through a wall with windows

One side of the wall in the building shown is 50 m wide and 170 m tall. On the wall, there are 552 identical windows, each of area 8.7 m^2. If the temperature difference between the interior and the exterior of the building is 8 °C, find the total rate of heat transfer by conduction through the wall with windows.



Given that the U-values of the window and the wall are 2.8 W m^{-2} K^{-1} and 4.2 W m^{-2} K^{-1} respectively.

Solution

Apply $\frac{Q}{t} = UA(T_H - T_C)$.

Rate of heat transfer by conduction through all windows
$= 2.8 \times (552 \times 8.7) \times 8 = 108\,000$ W

Rate of heat transfer by conduction through the wall (excluding windows)
$= 4.2 \times (50 \times 170 - 552 \times 8.7) \times 8 = 124\,000$ W

Total rate of heat transfer by conduction through the wall with windows
$= 108\,000 + 124\,000 = 232\,000$ W

Google Forms Microsoft Forms

Quick check 2

→ E-version in OUP Exercise Platform is also available.

1 Walls *P* and *Q* having the same area are made of the same material. Wall *P* is thicker than wall *Q*. Determine whether the following statements are true.

(a) Wall *P* has a lower thermal conductivity than wall *Q*. (T/F)

(b) Wall *P* has a lower U-value than wall *Q*. (T/F)

(c) Wall *P* conducts heat slower than wall *Q*. (T/F)

2 Which of the walls below has the best heat insulating property?

	Material	Thermal conductivity / W m^{-1} K^{-1}	Thickness / m
A	Brick	0.95	0.15
B	Concrete	1.10	0.2
C	Glass	1.05	0.05
D	Wood	0.14	0.1

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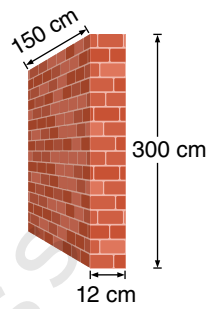
Learn by practice 2

1 The wall shown in the figure is made of brick. The temperatures at each side of the wall are 15 °C and 23 °C respectively. The thermal conductivity of the brick is 0.95 W m^{-1} K^{-1}.

(a) Find the U-value of the wall.

(b) Find the rate of heat flowing through the wall.

(c) Suggest a method to reduce the rate of heat flow through the wall.

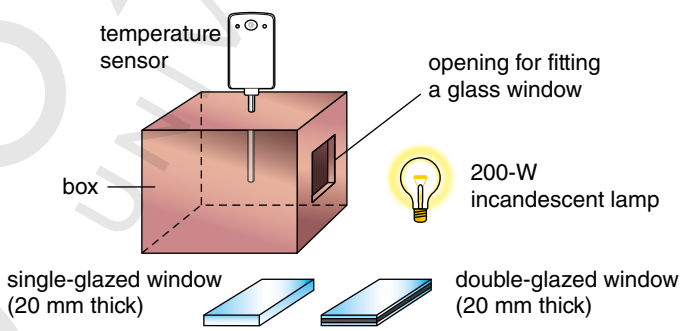


3 Heat transfer by radiation

Glass is a common material used in building envelopes. Apart from conduction, energy can also transfer through glass by radiation. Let us see the example below.

Example 5 Radiation through different windows

John sets up an experiment to record the temperature rise inside a box. The box has an opening for fitting a window. When it is fitted tightly with a window, the box is completely enclosed.



John first attaches a single-glazed window and then a double-glazed window to the opening. He finds that although the double-glazed window has a lower U-value, the rise in temperature in both cases is almost the same. Give an explanation for this result.

Solution

The U-value is related to heat transfer by conduction. In the experiment, heat is transferred through the windows predominantly by radiation. The amount of radiation passing through both windows is about the same. Therefore, the rise in temperature in both cases is almost the same.

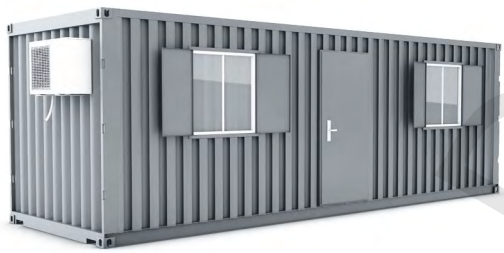
✎ Strictly speaking, the U-value corresponds to the heat transfer (including radiation) during indirect radiation exposure. In this set-up, energy is mainly transferred by radiation due to direct exposure to the light from the bulb. Therefore, the heat transfer through windows with different U-values is about the same.

☆ HKDSE 2017 Paper 2 Q.3(b)

→ This video demonstrates how this question is solved.

Video

DSE practice 1 | Container office



A shipping container is used as an office. The steel walls and roof have a total area of 52 m^2 and their U-value is 0.5 W m^{-2} K^{-1}. On a working day, the temperature of the exterior surface of the office is 32 °C and the temperature of the interior surface is maintained at 25 °C by an air conditioner. The total rate of heat transfer through the windows and the door is 772 W. The heat transfer through the floor is negligible. There are 4 people working inside the office and each person produces on average 120 J of thermal energy per second. The electrical equipment inside produces 80 J of thermal energy per second.

(a) Find the total rate of heat transfer through the building envelope. (2 marks)

(b) Estimate the minimum cooling capacity of the air conditioner. (2 marks)

(c) Explain why an air conditioner with the cooling capacity found in (b) is not enough to maintain the temperature of the interior surface at 25 °C on the following day

(i) if the temperature of the exterior surface of the office becomes 35 °C, or (2 marks)

(ii) if sunlight falls on the office and the air temperature outside is 32 °C. (3 marks)

Solution

(a) Total rate of heat transfer through the steel walls and roof

$= UA(T_H - T_C) = 0.5 \times 52 \times (32 - 25) = 182$ W 1M

Total rate of heat transfer through the building envelope

$= 182 + 772 = 954$ W 1A

(b) Minimum cooling capacity = minimum rate of heat removal

$= 954 + 4 \times 120 + 80$ 1M

$= 1514$ W 1A

(c) (i) The rate of heat transfer by conduction through the building envelope increases. 1A

Therefore, the heat needed to be removed per unit time by the air conditioner increases. 1A

(ii) The exterior surface absorbs heat by radiation 1A

and has a temperature higher than 32 °C. 1A

Besides, heat transfer into the container through the window by radiation. 1A

Therefore, the heat needed to be removed per unit time by the air conditioner increases.

Common mistake
Students may incorrectly state 'heat lost to surroundings' as the reason and not understand that heat is transferred from the surroundings into the office.

Common mistake
Students may not realize the difference between the surface temperature and the air temperature. Radiation does not affect the rate of heat transfer by conduction directly. It increases T_H, which in turn increases the rate of heat transfer by conduction.

Revision 3 Q27 (p.129)

DSE goal

Define OTTV and solve related problems.

DSE exam

12(2)Q3.4, 16(2)Q3.8, 21(2)Q3.4, 23(2)Q3.3

4 Overall Thermal Transfer Value (OTTV)

The total rate of heat flows into a building may change as the outdoor conditions change. In order to estimate the energy efficiency of a building, the **Overall Thermal Transfer Value**, or **OTTV** for short, is used. It is defined as the **average** rate of heat gain **per unit area** into a building through the building envelope by both **conduction and radiation** (Fig 3.1f).

$$\text{OTTV} = \frac{\text{average rate of heat gain}}{\text{total area}}$$

average rate of heat gain per unit area (W m^{-2})

$$\text{OTTV} = \left(\frac{Q_C}{t} + \frac{Q_S}{t}\right) \div A_T$$

total area of building envelope (m^2)

average rate of heat gain by conduction (W)

average rate of heat gain by radiation (W)

In calculating OTTV, we assume that the envelope of a building is **completely enclosed**. The heat gain by convection is negligible.

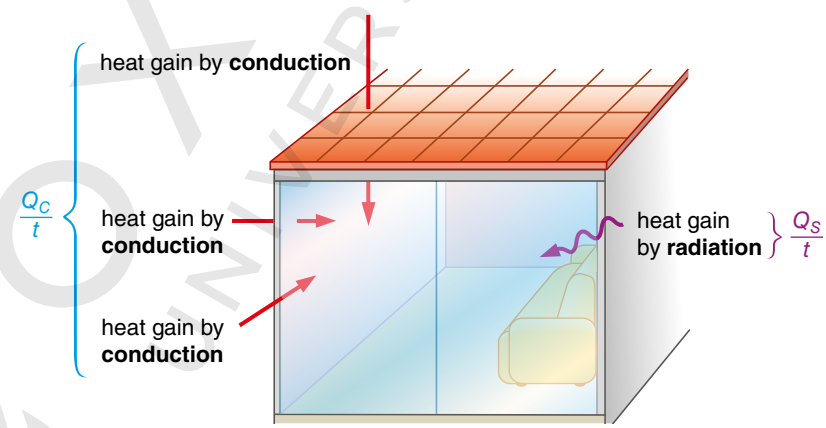


Fig 3.1f *Heat gain through a building envelope by conduction and radiation.*

In calculating $\frac{Q_C}{t}$, we do not use the instantaneous value of the temperature difference, but a value called the *equivalent temperature difference*. This value accounts for the temperature variations across the seasons and other factors affecting heat transfer (e.g. the orientation of a wall).

The smaller the OTTV of a building, the lower the rate of heat gain through the building envelope. Therefore, the electricity consumption for air conditioning can be reduced. Such a building has a higher energy efficiency.

Overall Thermal Transfer Value (OTTV) 總熱傳送值　　equivalent temperature difference 等效温差

Example 6 OTTV of a building

A building has four walls and a roof. There are some double-glazed windows on the walls. The total area of the windows is 700 m^2. The table shows the properties of the walls and the roof.

	Walls (excluding windows)	Roof
Total area / m^2	4000	400
U-value / W m^{-2} K^{-1}	0.6	0.4

(a) The equivalent temperature difference between the interior and the exterior of the building is 5 °C. Find the total rate of heat gain through the walls and the roof.

(b) The rate of heat gain per unit area through the windows is 130 W m^{-2}. Find the total rate of heat gain through the windows.

(c) Find the OTTV of the building.

Solution

(a) By $\frac{Q}{t} = UA(T_H - T_C)$,

total rate of heat gain through walls and roof

$= 0.6 \times 4000 \times 5 + 0.4 \times 400 \times 5 = 12\ 800$ W

(b) Total rate of heat gain through the windows

$= 130 \times 700 = 91\ 000$ W

(c) $\text{OTTV} = \frac{\text{average rate of heat gain}}{\text{total area}} = \frac{12\ 800 + 91\ 000}{4000 + 400 + 700} = 20.4$ W m^{-2}

In reality, the rates of heat transfer through walls and windows of different orientations are calculated separately as parameters such as the equivalent temperature difference and the solar factor are different.

In Hong Kong, the rate of heat gain by radiation through windows contributes a large portion of the OTTV of a building. The area and orientation of windows are, therefore, important factors in controlling the OTTV. In the next part, we will discuss the ways to lower the OTTV as well as improve the energy efficiency of a building.

Know more

OTTV of buildings in Hong Kong

OTTV is mainly used in engineering for describing energy efficiency. Its definition varies from place to place so as to cope with the local situation.

In Hong Kong, the heat gain by conduction through windows is neglected because this is usually small in comparison to heat gain by radiation. However, we may not exclude this factor in this book.

According to regulations, the OTTV should not exceed 21 W m^{-2} for a building tower.

The heat transfer through the ground is also neglected in calculating OTTV.

Google Forms Microsoft Forms

Quick check 3

→ E-version in OUP Exercise Platform is also available.

1 In summer, the outdoor temperature decreases at night. How does this affect the following quantities of a building?

(a) The OTTV

(b) The total rate of heat transfer through the building envelope

2 Two concrete buildings *X* and *Y* have the same dimensions and orientation. The total area of windows of *X* is larger than that of *Y*. Which building has a lower OTTV?

Learn by practice 3

1 The door and windows of a house are made of the same glass (Fig a). Their total area is 30 m^2. Their equivalent temperature difference between the interior and the exterior of the house is 8 °C.

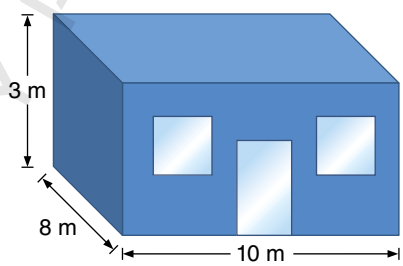


Fig a

Given:

U-value of glass = 5.6 W m^{-2} K^{-1}

U-value of the walls = 0.81 W m^{-2} K^{-1}

(a) Find the rate of heat gain by conduction through

(i) the walls,

(ii) the door and windows.

(b) The rate of heat gain per unit area through the glass by radiation is 100 W m^{-2}. Find the total rate of heat gain by radiation through the door and the windows.

(c) Find the OTTV of the house.

2 Consider the house shown in Figure b. The indoor temperature is maintained at 25 °C by an air conditioner. The average rate of heat transfer due to conduction through the walls and roof of the house is 1200 W and the rate of heat transfer through the window is 200 W.

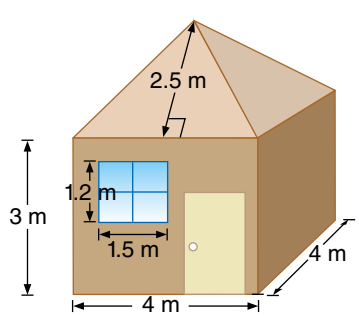


Fig b

(a) Estimate the OTTV of the house.

(b) If two people live in the house and each of them gives out thermal energy of 100 J each second, estimate the minimum cooling capacity of the air conditioner.

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DSE goal

Discuss factors affecting the OTTV.

Discuss the use of solar control window film in a building.

Discuss the factors affecting the energy efficiency of buildings.

DSE exam

13(2)Q3.6, 14(2)Q3.4, 15(2)Q3.5, 17(2)Q3.4, 18(2)Q3.6, 19(2)Q3.6

Simulation

→ This simulation demonstrates different ways in reducing heat transfer through a building envelope. It is available on OUP web in PC version (.exe) only.

5 Improving energy efficiency of buildings

The efficient use of energy in buildings can help reduce the overall impact on the environment. A well-designed building, which has a good thermal insulation and also allows suitable amount of daylight to enter the building for illumination, is therefore essential.

a Walls and roofs

A wall built out of a single material usually cannot provide strong support and enhance thermal insulation at the same time. However, a wall composed of more than one layer of different materials can serve the two purposes (Fig 3.1g).



Fig 3.1g *A wall composed of more than one layer to reduce heat transfer.*

Painting the outer surface of a building envelope with a light colour (Fig 3.1h) and building a roof-top garden (Fig 3.1i) can also lower the temperature of the outer surface.

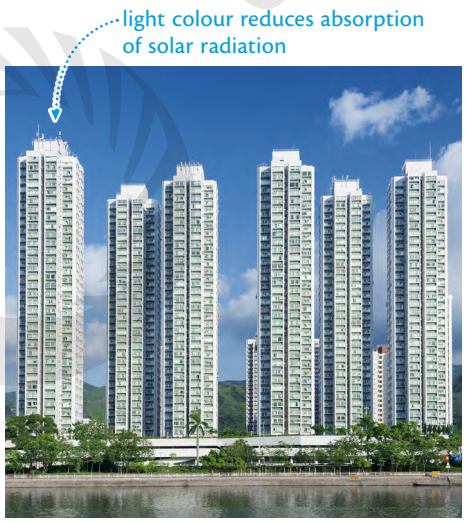


Fig 3.1h *Buildings with the outer surface in a light colour.*

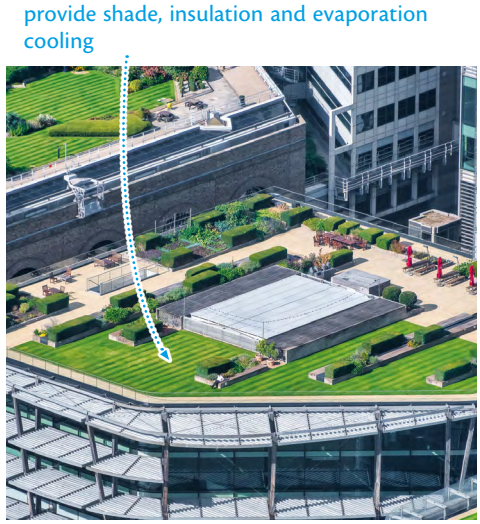


Fig 3.1i *Roof-top garden.*

b Windows

i Reducing heat transfer by conduction

The layer of air is thin to minimize convection.

A double-glazed window consists of two panes of glass with air (or another gas) between the panes (Fig 3.1j). Since air has a much lower thermal conductivity than glass, a double-glazed window can reduce the **heat transfer by conduction**. Some windows are even triple-glazed. They can further reduce the heat transfer by conduction.



Fig 3.1j Cutaway of double-glazed and triple-glazed windows.

Experiment 3b

Rates of heat transfer by conduction through windows (p.194)

Video

→ This video demonstrates Experiment 3b.

ii Reducing heat transfer by solar radiation

Under direct exposure to sunlight, heat is mostly transferred through windows by radiation. The **heat transfer by solar radiation** can be reduced by the following methods.

1. Reducing the amount of sunlight shining on the window
2. Reducing the area of the window
3. Allowing only a small proportion of sunlight to pass through the window

The following are examples of some types of glass that can reduce heat gain due to solar radiation (Fig 3.1k). They can either absorb or reflect the infra-red radiation that heat up the building.

(i) *(ii)* *(iii)*

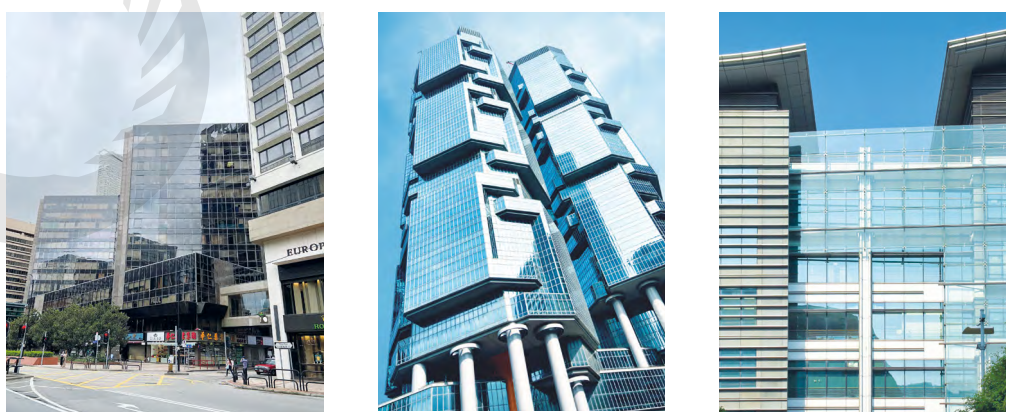


Fig 3.1k Buildings with (i) tinted glass; (ii) reflective glass and (iii) low-e glass.

Experiment 3c

Rates of heat transfer by radiation through windows (p.197)

Video

→ This video demonstrates Experiment 3c.

We can also install a **solar control window film** (Fig 3.1l) to an ordinary window (Fig 3.1m) to reduce heat gain due to radiation. The film can be easily installed. It is a low cost and convenient option to improve energy efficiency without renovation or structural work.



Fig 3.1l *Difference between the glass before and after installing solar control window film.*



Fig 3.1m *Applying solar control window film to an ordinary window.*

Among the above products, the low-e glass and the solar control window film perform better in improving the energy efficiency of a building. They **reflect most IR and UV radiation** while **allowing most of the visible light entering the building**. Table 3.1b compares the properties of different types of glass and window film.

Recall that IR is infra-red and UV is ultra-violet. Solar heat is mainly due to IR radiation. Blocking UV radiation can also prevent us from skin damage.

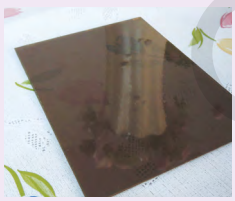
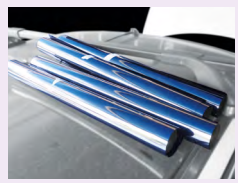
	Tinted glass	Reflective glass	Low-e glass	Solar control window film
				
Nature	coloured glass	glass coated with a reflective film	glass with low emissivity coating	thin film which is installed on a glass surface
Effect	absorbs all types of solar radiation	reflects all types of solar radiation	mainly reflects IR and UV radiation	mainly reflects IR and UV radiation
Effectiveness on reducing heat gain	not very effective as half of the energy absorbed by the glass is re-emitted into the interior	effective	effective	effective
Amount of visible light entering the building	reduced	reduced	slightly reduced	slightly reduced

Table 3.1b *Comparison between different types of glass and window film.*

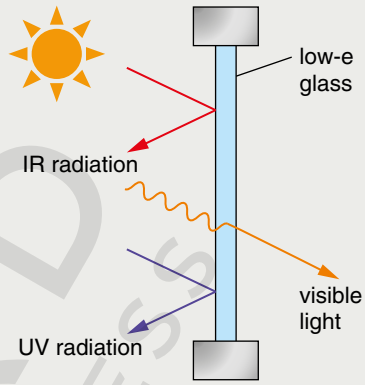
solar control window film 太陽隔熱膜

Know more

Low-emissivity glass

Low-emissivity (or low-e in short) glass is a glass with a thin layer of metallic coating (low emissivity coating) on it. The coating is much thinner than a human hair. It reflects IR and UV radiation. During summer, the heat outdoors is reflected to reduce heat from entering the building. During winter, the heat is trapped inside to keep the building warm.

On the other hand, the glass allows visible light to enter the building for natural lighting.



Insulating windows in winter

In some places, people stick pieces of bubble wrap on windows during winter to keep the room warm. The bubble wrap is made of plastics and air exists inside the bubbles. Since both plastics and air have low thermal conductivity, they reduce heat flow from the room to the outside.



C Natural lighting

Apart from windows, daylight can also enter a building from *skylights* (Fig 3.1n). Natural lighting can reduce the use of artificial lighting, hence reducing electricity consumption.

However, daylight may increase the demand of air conditioning owing to the strong solar heating effect. Quite often, an integrated approach, such as using skylights together with solar control window films, is used to optimize the energy efficiency.

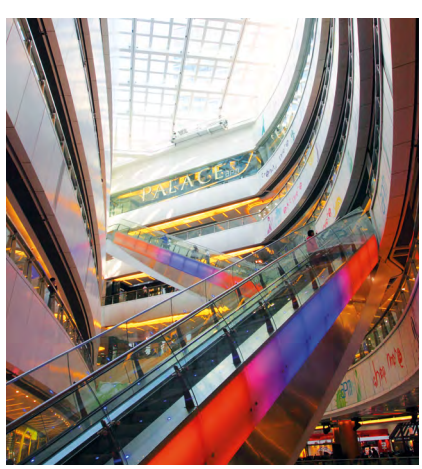


Fig 3.1n *Skylight in a shopping mall.*

skylight 天窗

d Sun-shading devices

Sun-shading devices (Fig 3.1o) can be installed on the top or at the sides of windows. They reduce the amount of sunlight falling directly on the windows. Well-designed sun-shading devices can help reduce the electricity consumption for air conditioning, but still allow a high level of natural lighting.



Fig 3.1o *Sun-shading devices of a building in the Hong Kong Science Park.*

e Orientation and shape of buildings

The orientation of a building greatly affects the amount of solar radiation entering it. For example, in the summer, a flat facing west is usually hotter than one facing south because it receives sunshine for a longer period of time.

The shape of a building can also help save energy. A carefully designed shape can enhance ventilation, hence reducing the electricity consumption for air conditioning (Fig 3.1p).

This is the Garden by the Bay in Singapore. This design can help the dome maintains a comfortable temperature.



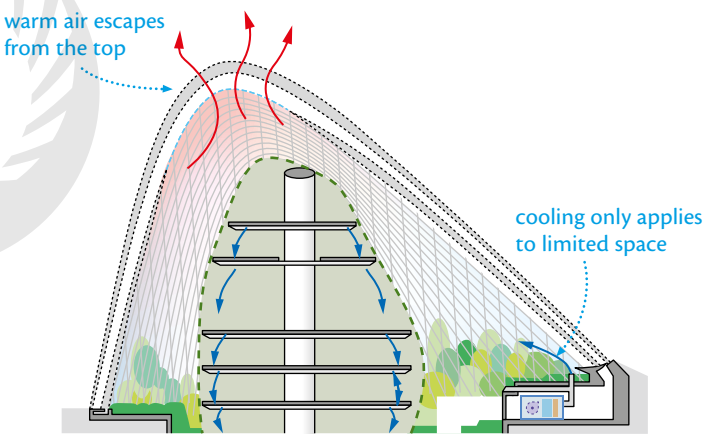


Fig 3.1p *The shape of the building helps maintain a comfortable temperature inside.*

Table 3.1c summarizes some ways that can improve the energy efficiency of a building.

Ways to lower the OTTV		Other ways to improve energy efficiency
Reduce heat transfer by conduction	**Reduce heat transfer by radiation**	
• Use building materials with lower U-values ($U\downarrow$) • Use light-coloured paint for outer surfaces ($\Delta T\downarrow$) • Build a roof-top garden ($\Delta T\downarrow$) • Use double-glazed windows ($U\downarrow$)	• Use low-e glass or solar control window films • Use sun-shading devices • Choose suitable orientation	• Use natural lighting • Use carefully designed shape to enhance ventilation

Table 3.1c *Ways to improve the energy efficiency of a building.*

Google Forms Microsoft Forms

Quick check 4

→ E-version in OUP Exercise Platform is also available.

1 To improve the energy efficiency of a building, which of the following EM waves should be prevented from transmitting through the windows? Put a tick in the corresponding box(es).

☐ Visible light

☐ Infra-red radiation

2 Which of the following can reduce heat transfer by conduction through a building envelope? Put a tick in the corresponding box(es).

☐ Install solar control window films on windows.

☐ Make good use of natural lighting.

☐ Use materials with low thermal conductivity for the building envelope.

Exercise 3.1

1 A wall is made of a single material. Its conductivity depends on its

A thickness. B area.

C material. D colour.

★**2** Which of the following factors affect(s) the U-value of a building component?

(1) The temperature difference between the two ends

(2) The thickness

(3) The area

A (2) only B (1) and (3) only

C (2) and (3) only D (1), (2) and (3)

★**3** Two rods, X and Y, are made of the same material (Fig a). The diameter and the length of X are d and $2L$ respectively. The diameter and length of Y are $2d$ and L respectively. The temperature difference across the two ends of X is the same as that of Y. If the rate of thermal conduction through X is R, what is the rate of thermal conduction through Y?

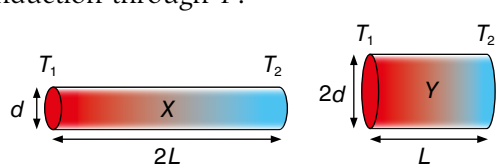


Fig a

A $\frac{R}{8}$ B $\frac{R}{4}$

C $4R$ D $8R$

★**4** Which of the following statements about the OTTV of a building envelope is/are correct?

(1) It is the total amount of heat transfer through the building envelope in one day.

(2) It varies with the temperatures inside and outside the building.

(3) It is affected by the orientation of the windows in the building.

A (3) only

B (1) and (2) only

C (2) and (3) only

D (1), (2) and (3)

★**5** Which of the following building components gives the best heat insulation?

	Thermal conductivity / W m^{-1} K^{-1}	Thickness / cm
A	0.1	4
B	1.1	0.6
C	0.6	20
D	0.04	5

★**6** Consider the building shown in Figure b. The two windows and the door are made of a single pane of glass of the same thickness.

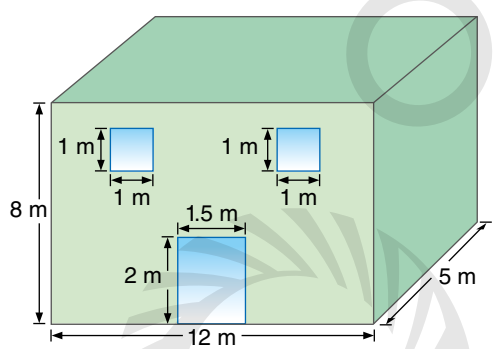


Fig b

The U-value of the walls and the roof is 2.4 W m^{-2} K^{-1}. The average rate of heat transfer per unit area through the windows and the door is 168 W m^{-2}. The equivalent temperature difference between the interior and exterior of the building is 6.68 °C. Estimate the OTTV of the building.

★**7** A cooling box made of foam is filled with 5 kg of ice at 0 °C and closed (Fig c). The temperature outside the box is 33 °C. The U-value of foam is 2 W m^{-2} K^{-1}.

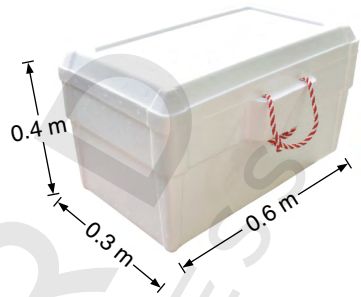


Fig c

(a) Estimate the total rate of heat exchange between the inside and the outside of the box.

(b) State the direction of heat flow.

(c) Suppose the temperature of the environment maintains at 33 °C. Estimate the time required for the ice to melt completely.

★**8** A double-glazed window of dimensions 80 cm × 60 cm consists of two glass layers of thickness 3 mm and an air gap of thickness 5 mm (Fig d).

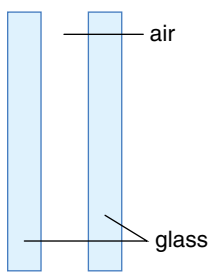


Fig d

Given:

thermal conductivity of glass = 1.05 W m^{-1} K^{-1}

thermal conductivity of air = 0.025 W m^{-1} K^{-1}

(a) Find the U-value of the double-glazed window.

(b) The temperature difference between the two sides of the window is 8 °C. Find the rate of heat transfer by conduction through the window.

★**9** Explain the difference(s) in how a tinted glass and a low-e glass reduce the heat transfer through them.

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3.2 Energy efficiency in transportation

About 31% of Hong Kong's annual energy usage is due to transportation. Most vehicles are powered solely by fossil fuels. However, vehicles solely or partly powered by electricity are getting more and more popular these days (Fig 3.2a). These vehicles are more energy efficient than fossil-fuel vehicles. Let us study how these vehicles help cut down the energy consumption in this section.

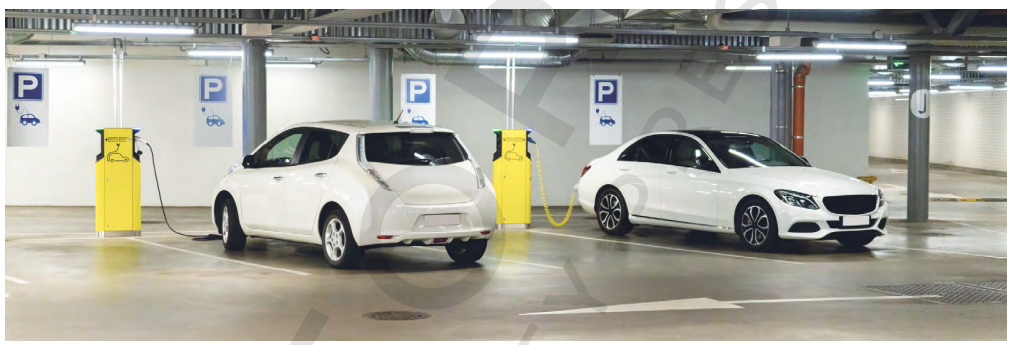


Fig 3.2a *Charging vehicles.*

1 Vehicles powered by different energy sources

a Fossil-fuel vehicles

A fossil-fuel vehicle uses an ***internal combustion engine*** for propulsion. In the engine, fuels burn and explode to power the vehicle. Much of the energy is wasted in heating up the vehicle and the air around it (i.e. turned into internal energy) during the process.

On the other hand, the vehicle brakes by making use of friction (Fig 3.2b). All the kinetic energy of the vehicle is wasted and becomes internal energy and sound energy (Fig 3.2c).

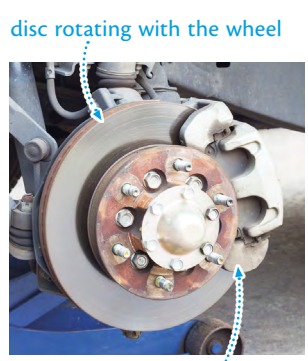


Fig 3.2b *Frictional braking.*

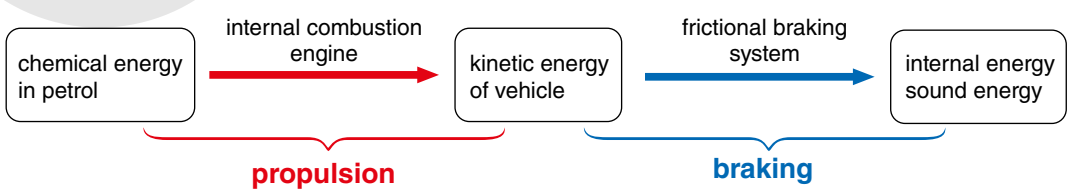


Fig 3.2c *Energy conversions in a fossil-fuel vehicle.*

internal combustion engine 內燃機

DSE goal

State the main components of the power system of electric vehicles.

Discuss the use of electric vehicles.

DSE exam

16(2)Q3.3, 18(2)Q3.1, 19(2)Q3.2, 19(2)Q3.3, 20(2)Q3.3, 21(2)Q.3, 22(2)Q3.7, 23(2)Q3.5, etc.

b Electric vehicles

In an **electric vehicle** (EV), rechargeable batteries converts chemical energy into electrical energy. Then the electric motor propels the vehicle by converting the electrical energy into the kinetic energy of the vehicle.

To reduce wasting energy during braking, an electric vehicle employs a **regenerative braking system**. During braking, the motor acts as a generator. The generator converts the kinetic energy into electrical energy, which is then stored in the batteries as chemical energy for later use (Fig 3.2d). In this way, the vehicle loses kinetic energy and slows down.

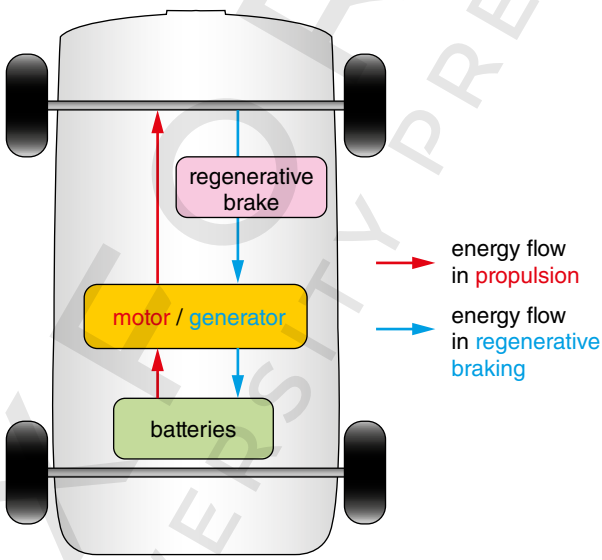


Fig 3.2d *Power system and regenerative braking system of an electric vehicle.*

Figure 3.2e shows the energy conversions in an electric vehicle during propulsion and braking.

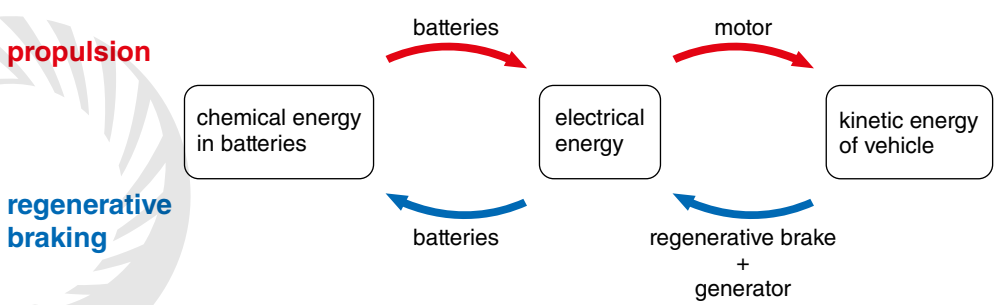


Fig 3.2e *Energy conversions in an electric vehicle.*

During braking, a regenerative braking system converts the kinetic energy of a vehicle into electrical energy, which is then stored as chemical energy by the batteries.

Experiment 3d

A model of regenerative braking system (p.201)

Video

→ This video demonstrates Experiment 3d.

electric vehicle 電動車　　regenerative braking system 再生制動系統

Cross-link

You learned how KE becomes electrical energy in Book 4 Ch 7.

The induced e.m.f. depends on the rate of flux cutting. When the vehicle brakes at a higher speed, the rate of flux cutting in the regenerative brake is higher, and so the rate of converting KE into electrical energy is higher.

The regenerative braking system recharges the batteries only when braking is applied, and the system cannot convert the KE of the vehicle into electrical energy effectively when the vehicle brakes at a very low speed. Therefore, the effectiveness of the system in saving energy varies on different driving conditions.

1 In a city with smooth traffic regulated by traffic lights (Fig 3.2f) or on a long downhill road, braking is often needed and the large KE of the vehicle can be utilized effectively.

2 On a highway, braking is seldom needed, so the system is seldom used.



Fig 3.2f *The regenerative braking system works when an electric vehicle brakes in front of a traffic light.*

3 In traffic congestion, although braking is frequently needed, the speed of the vehicle is low and the effectiveness in converting KE into electrical energy is low.

A frictional brake can lock the vehicle in position on a slope but a regenerative brake cannot.

Since the regenerative braking system is not effective in braking when the vehicle is slow, the vehicle **still needs a frictional braking system**.

Electric vehicles have the following advantages and disadvantages compared with fossil-fuel vehicles.

Advantages

A fossil fuel power plant emits air pollutants and greenhouse gases.

1 It does not directly emit air pollutants and greenhouse gases, thereby improving the air quality in urban area.

2 An electric vehicle is more energy efficient than a fossil-fuel vehicle. This is because:

- an electric motor is much more efficient in energy conversion than a combustion engine;
- some kinetic energy is recovered by the regenerative braking system;

On the other hand, the combustion engine in a fossil-fuel vehicle is kept running and consumes energy even the vehicle stops moving.

- the motor does not consume energy when the vehicle is stopped for a short time (e.g. while waiting for a green traffic light or in a traffic congestion).

3 The electricity cost is lower than the fuel cost of a fossil-fuel vehicle travelling the same distance.

4 It is very quiet.

Disadvantages

1 An electric vehicle is generally more expensive than a fossil-fuel vehicle due to the high cost of batteries.

2 The batteries are heavy and bulky (Fig 3.2g).

3 It takes much longer time to charge the batteries than refilling petrol.

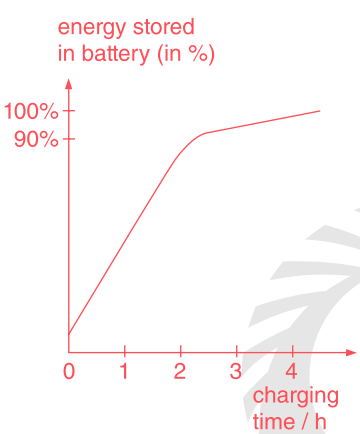
4 Charging the batteries could be inconvenient as the number of charging points in Hong Kong is limited.



Fig 3.2g *The battery in an electric bus.*

Some electric vehicles have short *ranges* (the maximum distance a vehicle can travel after fully charged or fuelled) due to the limited capacity of the batteries. However, some newer models can have a range similar to fossil-fuel vehicles.

A typical charging curve of a battery pack in an electric vehicle:



Tip

In equation $E = VIt$, if E is measured in W h, the unit of t should be h.

Example 7 Charging a battery pack of an electric vehicle

An electric vehicle uses a battery pack of capacity 24 kW h. The driver wants to charge the battery pack from 0% to 80% of its full capacity. If a medium charger is used, the battery pack will be charged with a terminal voltage of 220 V and an average current of 32 A. How long will the charging process take? Assume no energy is lost during the charging process.

Solution

By $E = VIt$,

$$t = \frac{E}{VI}$$

$$= \frac{24 \times 10^3 \times 80\%}{220 \times 32}$$

$$= 2.73 \text{ h}$$

The charging process takes 2.73 hours.

range 續行里程

Example 8 Battery capacity and range of an electric vehicle

An electric vehicle uses a battery pack of capacity 60 kW h. The power consumption of the vehicle motor is 4 kW when the vehicle moves at a constant speed of 30 km h^{-1}.

(a) Due to the energy loss from the electric circuit, the total energy supplied to the motor is only 85% of the energy stored in the battery. Estimate the range of the vehicle under this mode of driving.

(b) What is most of the energy output of the motor used for?

Solution

(a) Total energy output of motor = $60 \times 85\% = 51$ kW h

By $E = Pt$,

time for consuming all the energy = $\frac{E}{P} = \frac{51}{4} = 12.75$ h

Range of the vehicle = $vt = 30 \times 12.75 = 383$ km

The range of an average fossil-fuel vehicle is about 500 km. ▶

(b) Most of the energy output of the motor is used to overcome friction and air resistance.

Know more

Electric vehicles in Hong Kong

To promote the usage of electric vehicles, the Hong Kong government provides many tax benefits. According to the data obtained from EPD website in April 2024, the number of electric vehicles has soared from about 180 in 2010 to about 87 900 in March 2024, representing about 9.7% of the total number of vehicles. The first electric single-deck bus and first electric double-deck bus were launched in 2015 and 2022 respectively. As of February 2024, there were 76 electric buses.

As of the end of March 2024, there were more than 8000 charging stations for public use, including about 1400 quick chargers, which can charge a typical electric vehicle to 80% in about 30 minutes.



The first electric double-deck bus in Hong Kong.

According to the Hong Kong Roadmap on Popularisation of Electric Vehicles (the EV Roadmap) announced in March 2021, the government aims to stop the registration of new fuel-powered private cars, including hybrid cars, by 2035 or earlier. By then, all newly registered private cars in Hong Kong will be pure electric vehicles.

Google Forms Microsoft Forms

Quick check 5

→ E-version in OUP Exercise Platform is also available.

1 In which of the following driving modes does the regenerative braking system of an electric vehicle have the highest effectiveness?

A Driving uphill

B Driving in a city with traffic lights

C Driving on a highway with smooth traffic

D Being stuck in traffic

2 *True or False*:

(a) Electrical vehicles have a higher energy efficiency than fossil-fuel vehicles. (T/F)

(b) The regenerative braking system converts all kinetic energy of the vehicle back to the battery during braking. (T/F)

(c) Regenerative braking helps increase the end-use energy efficiency of a vehicle. (T/F)

(d) Frictional braking converts kinetic energy into internal energy. (T/F)

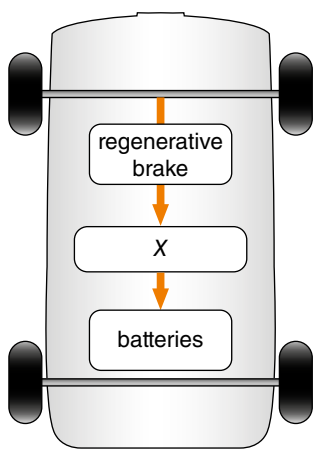
(e) Frictional braking is required when an electric vehicle brakes from a low speed. (T/F)

(f) Electric vehicles improve air quality in urban area. (T/F)

3 The figure on the right shows an electric vehicle. The arrows show how energy flows when the vehicle is braking.

(a) Does *X* act as a motor or a generator?

(b) State the energy conversion when the vehicle is braking.



Learn by practice 4

1 A completely discharged electric vehicle of capacity 75 kW h is being charged with an input voltage of 220 V. If the average current of charging is 40 A, estimate the time required, in hours, for the electric vehicle to be fully charged.

2 When an electric car travels steadily at 50 km h^{-1}, the power consumed by the motor is 5 kW and the range of the car is about 450 km. The energy supplied to the motor is 90% of that stored in the batteries. Estimate the capacity (in kW h) of the batteries.

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Example 9 Running costs of cars

A customer is choosing between a fossil-fuel car and an electric car.

(a) Suggest two reasons why an electric vehicle is more energy efficient than a fossil-fuel vehicle of similar size.

(b) The fuel consumption of the fossil-fuel car is 7.8 L per 100 km while the power consumption of the electric car is 14.3 kW h per 100 km. Each litre of fuel costs \$15 while each kW h of electricity costs \$2.9. Suppose the car has to runs an average of 60 km a day. What is the cost of fuel/electricity used by each car each day?

Solution

(a) An electric motor is more efficient than a combustion engine in terms of energy conversion.

An electric vehicle has a regenerative braking system which can convert the kinetic energy of the vehicle back into chemical energy in the battery for future use during braking.

(b) Fuel used by the fossil-fuel car each day

$= 7.8 \times \frac{60}{100} = 4.68$ L

Cost of fuel used each day

$= 15 \times 4.68 = \$70.2$

Electricity used by the electric car each day

$= 14.3 \times \frac{60}{100} = 8.58$ kW h

Cost of electricity used each day

$= 2.9 \times 8.58 = \$24.9$

▶ Exercise 3.2 Q9 (p.120)

✎ Hong Kong's first hydrogen fuel cell electric bus was in service in February 2024. The bus is a double-decker which has a capacity of 119. Each refueling takes about 10 to 15 minutes and the bus can run up to 400 km.

✎ Fuel-cell vehicles only emit water vapor and warm air when running, making them zero-carbon-emission vehicles. However, whether these vehicles are more environmental friendly remains controversial. As currently almost all commercial hydrogen is produced directly from fossil fuels, it is an industry with high carbon emissions.

Know more

Types of electric vehicles

There are different types of electric vehicles.

1 **Battery electric vehicles:** Batteries are used to provide electricity. They can be charged by connecting to a charger.
2 **Fuel-cell vehicles:** The vehicles are installed with fuel cells, which use the chemical reaction of hydrogen (as a fuel) and oxygen to generate electricity. Refilling a hydrogen cell takes only several minutes.
3 **Capacitor vehicles:** The vehicles use a *supercapacitor* to store electrical energy. The supercapacitor can only store a small amount of energy but can be charged very quickly.

supercapacitor 超級電容器

C Hybrid vehicles

DSE goal

State the main components of the power system of hybrid vehicles and compare their end-use energy efficiency to fossil-fuel vehicles.

DSE exam

14(2)Q3.6, 16(2)Q3.3, 18(2)Q3.1, 19(2)Q3.3

Flipped classroom

Watch a video and answer the questions.

Simulation

→ This simulation shows how the power system and braking system of a hybrid vehicle works. It is available on OUP web in PC version (.exe) only.

A **hybrid vehicle** is a combination of a fossil-fuel vehicle and an electric vehicle (Fig 3.2h). It can be propelled using the combustion engine, or the electric motor, or both together if more power is needed. A computer is used to control them to work in the most efficient way in different situations. Different types of hybrid vehicles may work differently. Table 3.2a shows an example.

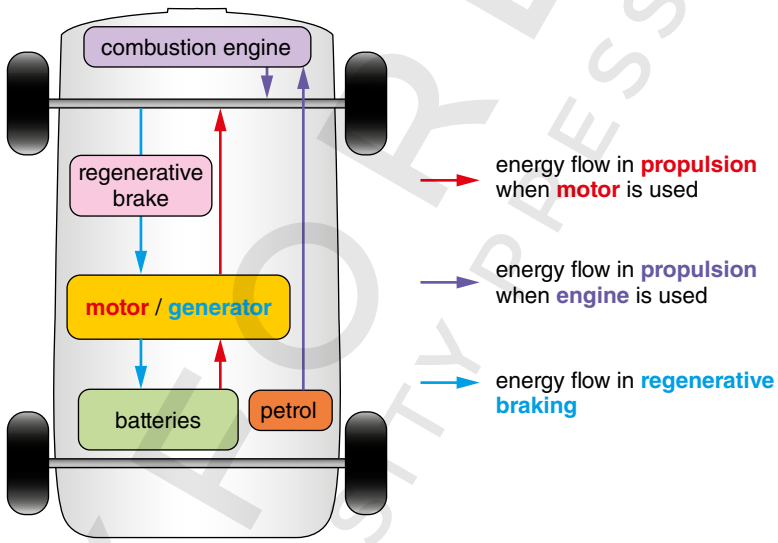


Fig 3.2h Power system and regenerative braking system of a hybrid vehicle.

Driving condition	Low-speed driving	Acceleration / going uphill	High-speed driving	Braking
Motor used	✓	✓	✓	✓ (acts as generator)
Engine used	✗	✓	✓	✗

Table 3.2a How a hybrid vehicle works in different situations.

Similar to an electric vehicle, a hybrid vehicle also has a regenerative braking system and a frictional brake. The batteries can be charged through regenerative braking and by the combustion engine.

▶ Some hybrid vehicles can be connected to mains electricity supply to recharge the batteries when the vehicle is not in use.

Figure 3.2i shows the energy conversions in a hybrid vehicle during propulsion and braking.

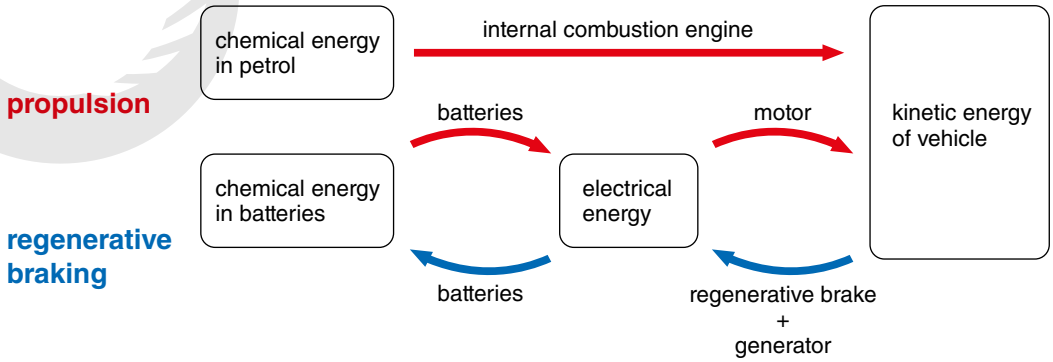


Fig 3.2i Energy conversions in a hybrid vehicle.

hybrid vehicle 混合動力車

A hybrid vehicle has the following advantages and disadvantages compared with a fossil-fuel vehicle.

Advantages

1 It is more energy efficient. The reasons include:

- some waste energy in braking is recovered by the regenerative braking system and reused later;
- the combustion engine can be turned off when the vehicle is not moving;
- a smaller combustion engine, which is more efficient than a bigger one, is used since the electric motor can share the load.

2 It uses less fuel, meaning that it emits fewer air pollutants and needs a lower fuel cost.

3 It is quiet when the combustion engine is not in use.

When a hybrid vehicle is stationary, its internal combustion engine turns off. When the vehicle starts to move, the electric motor spins the combustion engine first so that it runs smoothly before fuel flows through it again.

Disadvantages

1 It is expensive.

2 The maintenance cost is high.

When compared with an electric vehicle, the refuelling time of a hybrid vehicle is much shorter. However, a hybrid vehicle is less energy efficient and emits air pollutants.

This is because an electric motor is more efficient in energy conversion than a combustion engine.

Example 10 Charging battery directly by engine

In some types of hybrid vehicles, the combustion engine can propel the vehicle and charge the battery at the same time when the travelling speed is low.

(a) Explain why these vehicles can charge their batteries when their speeds are low.

(b) State the energy conversions for the charging process briefly.

Solution

(a) When the vehicle travels at a low speed, the power output required is low. The engine can then provide excess power to charge the battery.

(b) Chemical energy in petrol
→ KE of generator
→ electrical energy
→ chemical energy in battery

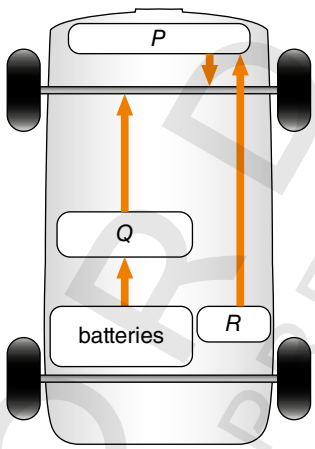
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Quick check 6

→ E-version in OUP Exercise Platform is also available.

1 The figure below shows the energy flow in a hybrid vehicle when the vehicle is accelerating with full power.



(a) Name *P* to *R*.

(b) Which part of the hybrid vehicle acts as a generator when the vehicle brakes?

2 *True or False:*

(a) The combustion engine in a hybrid vehicle is smaller than that in a fossil-fuel vehicle. (T/F)

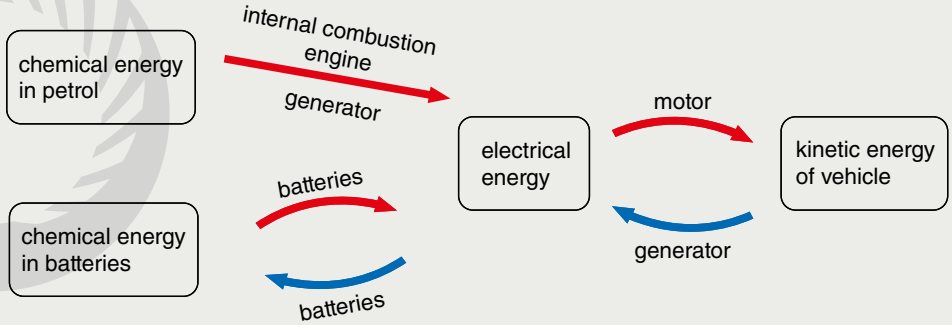
(b) The batteries in some hybrid vehicles can be charged by the combustion engine. (T/F)

(c) A hybrid vehicle does not emit air pollutant directly. (T/F)

Know more

Series hybrid vehicles

The hybrid vehicles discussed above are called parallel hybrid. There is another type of hybrid system called series hybrid. In this type of hybrid, the combustion engine is used for generating electricity and the vehicle is propelled by electric motor only.



Compared with the parallel hybrid, the series hybrid system can provide a higher combustion efficiency as the combustion engine works constantly at the optimal speed. Since the engine is not coupled to wheels, the engine can be set to run in the most efficient way (revolution per minute).

Table 3.2b compares some properties of the above three types of vehicles.

	Fossil-fuel vehicles	Electric vehicles	Hybrid vehicles
Power system	combustion engine (less efficient)	electric motor (more efficient)	combustion engine and electric motor (more efficient)
Power source	fossil fuel	batteries	fossil fuel and batteries
Braking system	frictional braking	regenerative braking and frictional braking	regenerative braking and frictional braking
Energy refilling	• convenient • short refuelling time	• insufficient charging points in some places • long recharging time	• convenient • short refuelling time
Major advantage	lower price	• reduce air pollution in urban area • energy efficient • very quiet • low electricity cost	• energy efficient • low fuel cost
Major disadvantage	• emit air pollutants • high fuel cost	• higher price • heavy and bulky batteries	• higher price • heavy and bulky batteries • emit air pollutants

Table 3.2b *Comparison between fossil-fuel, electric and hybrid vehicles.*

DSE exam
14(2)Q.3(c)

2 Effects of power plants

Using an electric motor to propel a vehicle is more efficient than using a combustion engine. However, to give a fair comparison between the vehicles, we should not ignore what happens in the production of electricity. Figure 3.2j shows schematically how an electric vehicle obtains energy from a coal-fired power plant.

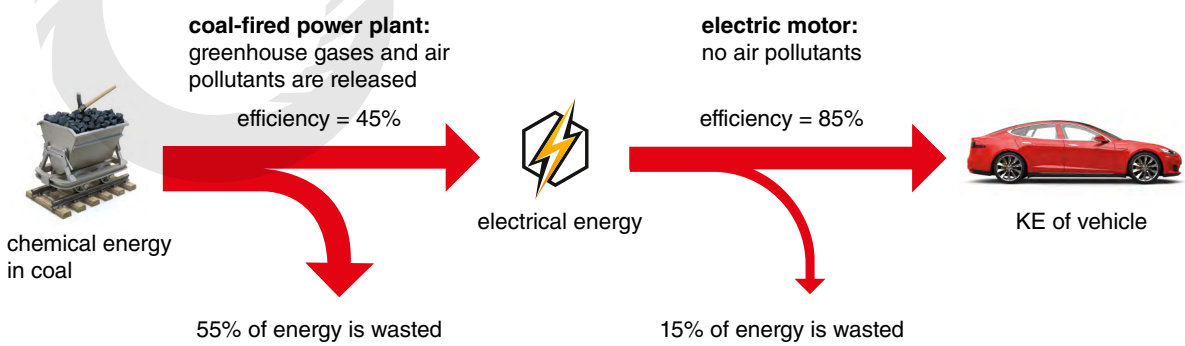


Fig 3.2j *How an electric vehicle obtains energy from a coal-fired power plant.*

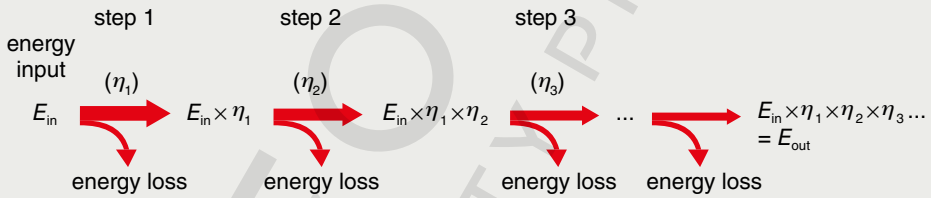
Although the energy efficiency of an electric motor can be quite high (~85%), the **overall energy efficiency** of running an electric vehicle with electricity generated from a coal-fired power plant can be only about 38% (= 45% × 85%), which is not significantly more efficient than a fossil-fuel vehicle.

Know more

Overall Energy Efficiency

As mentioned in Chapter 1, an energy conversion can never be perfect. It has an efficiency η that indicates the percentage of the energy input E_{in} turned into the useful energy output E_{out} (= $E_{in} \times \eta$).

For a process involves successive steps of energy conversion, each has its own efficiency $\eta_{n(n = 1, 2, 3...)}$,



The overall energy efficiency η_o of the process can be found by multiplying the efficiency of each step under consideration, i.e.

$$\eta_o = \frac{\text{useful energy output}}{\text{energy input}} \times 100\%$$
$$= \eta_1 \times \eta_2 \times \eta_3 \times \ldots$$

Moreover, although electric vehicles do not cause air pollution to the places they pass by, the coal-fired power plants do pollute the environment (Fig 3.2k).



Fig 3.2k *Air pollution caused by a coal-fired power plant.*

In the future, if clean energy like wind power and solar power become more popular, using electric vehicles would become more environmental-friendly.

DSE goal

Discuss the advantages of public transportation systems and give examples.

DSE exam

18(2)Q3.2

3 Efficient use of transportation

Different modes of transportation have different energy efficiencies. In general, a public transportation system can carry many passengers and is more energy efficient than private vehicles (Fig 3.2l). Among all types of vehicles, the electric train takes the smallest amount of energy to transport a passenger for 1 km.

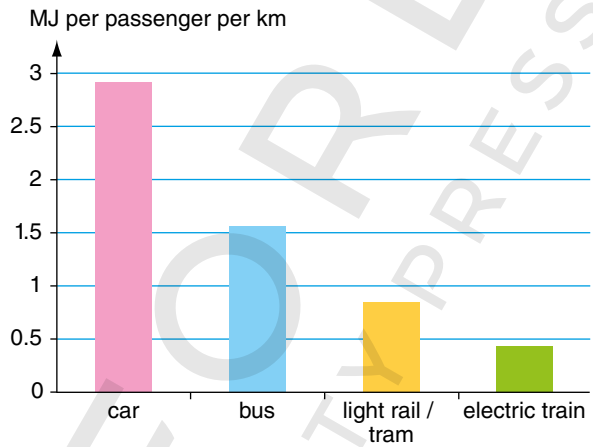


Fig 3.2l *Average energy consumption for different modes of transportation.*

A city may have the following benefits if its public transportation system is well designed.

1 The overall emission of carbon dioxide (Fig 3.2m) and smog-causing particles is reduced as less energy is used.

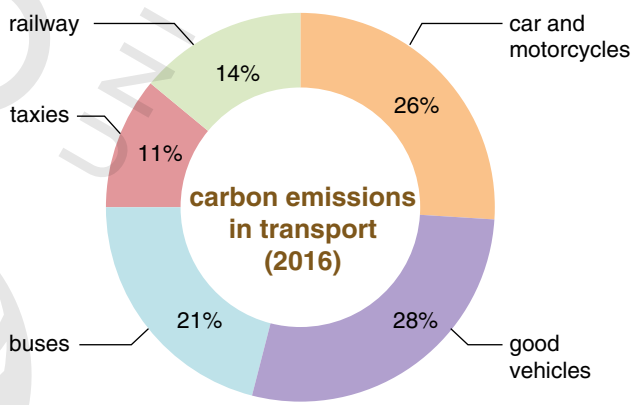


Fig 3.2m *Carbon emissions from different types of vehicles in Hong Kong in 2016. (Data obtained from HK2050isNOW website in Oct 2022)*

2 The air pollution in urban area due to car emissions is reduced.

3 Traffic congestion is reduced as there are fewer vehicles on the road.

In addition, if electric public transport is adopted, the problems of carbon emission and air pollution can further be improved. Examples of electric public transport include trams, trains and electric buses.

▶ Trams and trains do not carry batteries. Electricity is transmitted via cables.

Example 11 Energy consumed by buses and cars

In Hong Kong, about 3 million passengers travel by bus every day. Suppose that all these passengers travelled by taxi instead of bus.



(a) How much more energy would be used due to these passengers per kilometre of travel? The amount of energy needed per person per kilometre for buses and taxis is about 1.5×10^6 J and 2.9×10^6 J respectively.

(b) How much more carbon dioxide would be emitted due to these passengers per kilometre of travel? The amount of carbon dioxide emission associated with a person per kilometre for buses and taxis is 38 g and 120 g respectively.

Solution

(a) Additional amount of energy used per km
$= (2.9 - 1.5) \times 10^6 \times 3 \times 10^6 = 4.2 \times 10^{12}$ J

(b) Additional amount of carbon dioxide emitted per km
$= (120 - 38) \times 3 \times 10^6 = 2.46 \times 10^8$ g $= 2.46 \times 10^5$ kg

Google Forms Microsoft Forms

Quick check 7

→ E-version in OUP Exercise Platform is also available.

1 Which of the following methods of transportation in Hong Kong is the most energy efficient?

A Bus B Car
C Electric train D Tram

2 Which of the following are the advantages of using buses over private cars?

☐ Lower cost of transportation per passenger
☐ Shorter travelling time
☐ Less fuel consumption per passenger

3 *True or false:* The overall energy efficiency of an electric vehicle is lower than that of a fossil-fuel vehicle. (T/F)

Exercise 3.2

★**1** The carbon dioxide emission per passenger for a taxi and an electric train to travel 1 km is 220 g and 9 g respectively. On average, Cecilia travels 40 km every day. How much more carbon dioxide would be emitted in a year if Cecilia travelled by taxi instead of by electric train?

A 3080 kg B 3210 kg
C 8440 kg D 8800 kg

★**2** The battery of an electric vehicle can store at most 30 kW h of energy. The vehicle can travel for 300 km after it is fully charged. The cost of electricity is $2.9 per kW h. What is the cost of electricity to drive the vehicle for 10 km?

A $2.9 B $5.5
C $46.4 D $55.2

★**3** A fully discharged battery in an electric vehicle is charged at 220 V for 2 hours. The average charging current is 25 A.

(a) What is the amount of energy (in kW h) stored in the battery after it is charged?

(b) If the cost of electricity is $2.9 per kW h, what is the cost of this charging process?

★**4** Explain the benefits of building an electric train network in a highly populated city in terms of the emission of carbon dioxide.

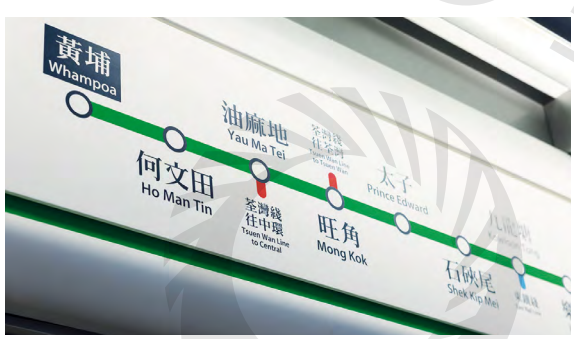


Fig a

★**5** A hybrid car can travel 20 km on one litre of petrol. The cost of petrol is $15 per litre.

(a) What is the fuel cost to drive the car for 50 km?

(b) State one advantage of driving a hybrid vehicle over an electric vehicle in Hong Kong.

★**6** A hybrid vehicle is moving with the maximum power output.

(a) Is carbon dioxide emitted? Explain your answer.

(b) Describe the energy conversions involved.

★**7** What should the following vehicle inside the airport be, a fossil-fuel vehicle or an electric vehicle? Why?



Fig b

★**8** To travel 100 km, a fossil-fuel car needs 8.3 L of petrol while a hybrid car needs only 6.3 L of petrol. Each litre of petrol costs $17.

(a) A customer buys the hybrid car. What is the total distance that the car must travel in order to save $10 000 in the cost of petrol compared to a fossil-fuel car?

(b) Besides lower fuel cost, state two other advantages of a hybrid car over a fossil-fuel car.

★**9** The motor of an electric vehicle consumes a power of 9000 W in order to travel at a constant speed of 100 km h^{-1}. The range of the car is about 400 km under this mode of driving. The energy supplied to the motor is 90% of the energy stored in the batteries.

(a) What is the capacity (in kW h) of its battery?

(b) State one advantage and one disadvantage of an electric vehicle over a fossil-fuel vehicle.

★**10** A car has a regenerative braking system with an overall energy efficiency of 35%. The mass of the car is 1500 kg. How much energy (in kW h) can be stored in the battery if the car decelerates from 90 km h^{-1} to 30 km h^{-1}?

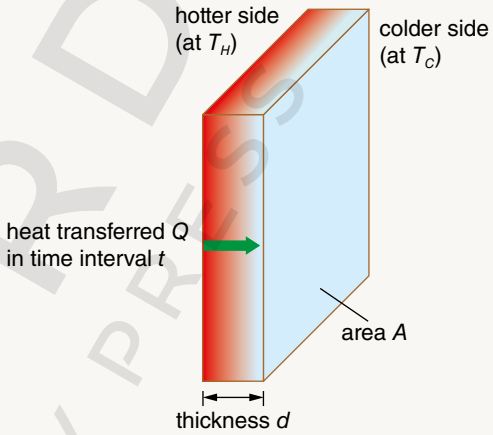
Key concepts 3

1 Energy performance in buildings

Rate of heat transfer by conduction

$$\frac{Q}{t} = \kappa\frac{A(T_H - T_C)}{d}$$

κ: thermal conductivity measured in W m^{-1} K^{-1}



Thermal transmittance / U-value (U)

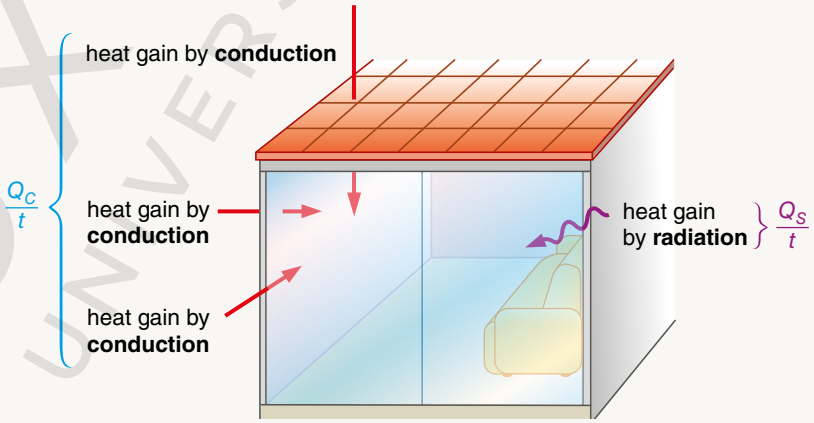
- For a single material, $U = \frac{\kappa}{d}$
- $\frac{Q}{t} = UA(T_H - T_C)$
- Unit: W m^{-2} K^{-1}

Overall Thermal Transfer Value (OTTV)

- Average rate of heat gain per unit area into a building through the building envelope by both conduction and radiation

$$\text{OTTV} = \left(\frac{Q_C}{t} + \frac{Q_S}{t}\right) \div A_T$$

- Unit: W m^{-2}



Improving energy efficiency of buildings

- Reducing heat transfer due to conduction (by using double-glazed windows, walls with thermal insulation, etc.)





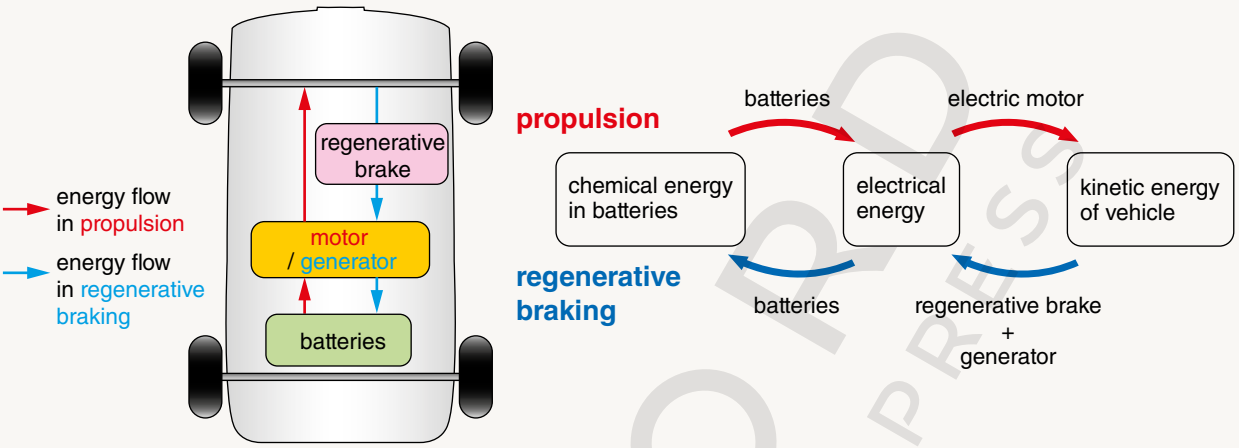


- Reducing heat transfer due to radiation (by using solar control window film, low-e glass, etc.)

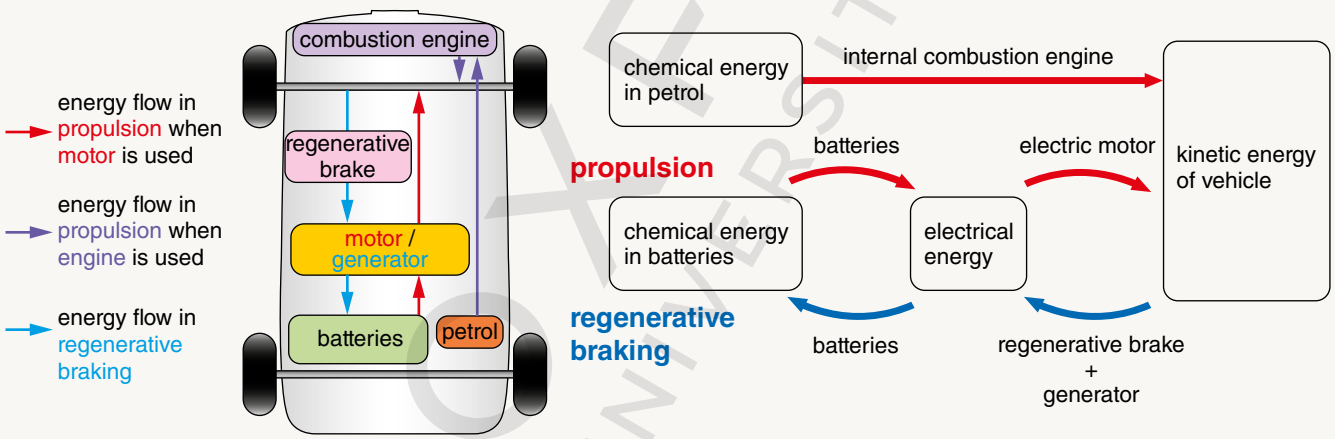
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2 Energy performance in transportation

Electric vehicles



Hybrid vehicles



Comparing different types of vehicles

	Fossil-fuel vehicles	Electric vehicles	Hybrid vehicles
Power system	combustion engine (less efficient)	electric motor (more efficient)	combustion engine and electric motor (more efficient)
Braking system	frictional braking	regenerative braking and frictional braking	regenerative braking and frictional braking
Advantage	• short refuelling time • lower price	• reduce air pollution in urban area • energy efficient • low electricity cost	• short refuelling time • energy efficient • low fuel cost
Disadvantage	• emit air pollutants • high fuel cost	• higher price • long recharging time • heavy and bulky batteries	• higher price • emit air pollutants • heavy and bulky batteries

Key terms

1 electric vehicle 電動車 p.107
2 hybrid car 混合動力車 p.113
3 internal combustion engine 內燃機 p.106
4 Overall Thermal Transfer Value (OTTV) 總熱傳送值 p.96
5 regenerative braking system 再生制動系統 p.107
6 solar control window film 太陽隔熱膜 p.101
7 thermal conductivity 導熱率 p.87
8 thermal transmittance (U-value) 熱傳送係數（U 值） p.90

Revision 3

Multiple-choice questions

★1 3.2 Travelling by bus is more energy efficient than travelling by car. Which of the following is/are the reason(s)?

(1) Buses can carry more passengers than a car.
(2) A bus uses less energy than a car in travelling the same distance.
(3) Buses have regenerative braking system.

A (1) only B (2) only
C (1) and (3) only D (2) and (3) only

★2 3.1



Fig a

Which of the following about installing a glass curtain wall (Fig a) instead of a concrete wall with windows in a building is correct?

(1) This can reduce the energy used in lighting.
(2) This can reduce the energy used in air conditioning.
(3) This can reduce the OTTV of the building.

A (1) only B (1) and (3) only
C (2) and (3) only D (1), (2) and (3)

★3 3.1 Which of the following statements about the thermal conductivity of a material is/are correct?

(1) It shows how well the material conducts heat.
(2) Its value is affected by the material's colour.
(3) Using materials of lower thermal conductivity to build the envelope can reduce its OTTV.

A (1) only B (1) and (3) only
C (2) and (3) only D (1), (2) and (3)

★4 3.1 A wall consists of two layers *X* and *Y*. Which of the following changes would increase the U-value of the wall?

(1) The temperature difference between the two sides of the wall increases.
(2) The thickness of layer *X* decreases.
(3) Add mineral wool between layers *X* and *Y*.

A (2) only B (3) only
C (1) and (2) only D (1) and (3) only

★5 3.1 Which of the following changes can reduce the rate of heat gain by a room through a wall?

(1) Paint the outer surface of the wall black.
(2) Cover the wall with mineral wool.
(3) Increase the thickness of the wall.

A (2) only B (1) and (3) only
C (2) and (3) only D (1), (2) and (3)

★6 3.2 The maximum amount of energy stored in the battery of an electric vehicle is 35 kW h. It takes 4 hours to charge the battery at 220 V from 20% to 100%. What is the average charging current?

A 3.64 A
B 14.5 A
C 31.8 A
D 49.7 A

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***7** 3.2 Which of the following statements about a hybrid car is/are correct?

(1) It does not emit air pollutants during travelling.

(2) Its combustion engine is more efficient than that of a fossil-fuel vehicle.

(3) Its electric motor can work together with the combustion engine.

A (1) only

B (2) only

C (1) and (3) only

D (2) and (3) only

****8** 3.1 Two walls X and Y are made of the same material and painted the same colour. Both walls have one surface facing east and the other one facing west (Fig b). The dimensions of X and Y are $2h \times d \times t$ and $h \times d \times 2t$ respectively.

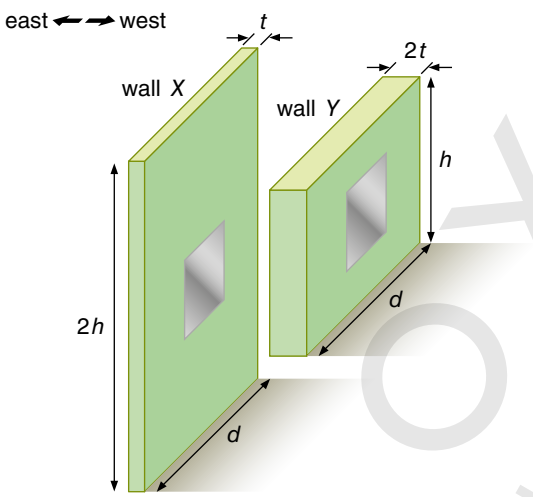


Fig b

A metal plate of negligible thickness is attached to each wall on the surface facing west. At first, the surfaces of the walls and the metal plates are at the same temperature. The sun then rises and shines on the surfaces of the walls facing east. Which metal plate increases to a higher temperature after a certain period of time?

A The one on wall X

B The one on wall Y

C Both are at the same temperature.

D Cannot be determined

Refer p.87

9 3.1 **HKDSE 2015 Paper 2 Q3.5**

Figure c shows the percentage of absorption of different electromagnetic radiations when passing through two types of glasses X and Y. One of them is to be chosen for windows of buildings in Hong Kong. With the consideration of energy efficiency of buildings, which statement is correct?

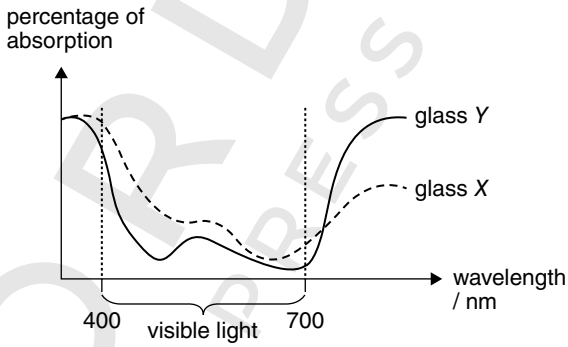


Fig c

A X should be chosen as it is better than Y in reducing the need of air conditioning and the amount of lighting.

B Y should be chosen as it is better than X in reducing the need of air conditioning and the amount of lighting. (28.08%)

C X should be chosen as it can greatly reduce the need of air conditioning although it needs a bit more amount of lighting than Y.

D Y should be chosen as it can greatly reduce the need of air conditioning although it needs a bit more amount of lighting than X.

10 3.1 **HKDSE 2015 Paper 2 Q3.6**

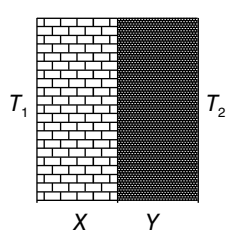


Fig d

Figure d shows a wall composed of two layers X and Y of equal thicknesses. The thermal conductivity of the material of X is higher than that of Y. The two sides of the wall are maintained at different temperatures T_1 and T_2.

Which of the following statements is/are correct?

(1) The thermal transmittance of layer *X* is higher than that of layer *Y*.

(2) The energy flowing through layer *X* per second is greater than that through layer *Y*.

(3) The temperature drop across layer *X* is smaller than that across layer *Y*.

A (1) only B (2) only

C (1) and (3) only D (2) and (3) only (28.42%)

11 HKDSE 2018 Paper 2 Q3.2
3.2

Carbon dioxide emission associated with a passenger travelling in a MTR train for 1 km is about 50 g while that for a car is over 200 g. The possible reason(s) is/are

(1) The MTR train and the car need energy to move the vehicle itself.

(2) The energy efficiency in transporting a passenger for 1 km is higher in the case of the MTR train.

(3) No burning of fossil-fuel is involved in the production of energy for the MTR train.

A (1) only B (2) only (57.9%)

C (1) and (3) only D (2) and (3) only

12 HKDSE 2019 Paper 2 Q3.5
3.1

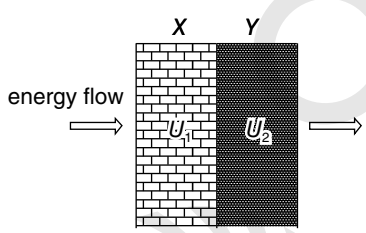


Fig e

A wall is composed of layers *X* and *Y* of U-values U_1 and U_2 respectively (Fig e). Both layers have the same thickness and dimensions, and there is no air gap between them. Which expression gives the U-value of the wall?

A $U_1 + U_2$

B $\frac{1}{2}(U_1 + U_2)$

C $\frac{2U_1U_2}{U_1 + U_2}$

D $\frac{U_1U_2}{U_1 + U_2}$ (26.55%)

13 HKDSE 2020 Paper 2 Q3.6
3.1

A room is kept cool by an air-conditioner of cooling capacity *P*. The temperature inside and outside the room are 27 °C and 31 °C respectively. The rate of heat flowing into the room by radiation through windows and that by conduction are in the ratio 1:4. If the cooling capacity is raised to 2*P* while the temperature outside the room is still 31 °C, estimate the temperature inside the room. Assume that the rate of heat flowing into the room by radiation is unchanged.

A 21 °C

B 22 °C (24.1%)

C 23 °C

D 25 °C

14 HKDSE 2021 Paper 2 Q3.4
3.1

Figure f shows a concrete building of dimensions 35 m × 40 m × 50 m. It is given that the Overall Thermal Transfer Value (OTTV) of a building should not exceed 24 W m^{-2}.

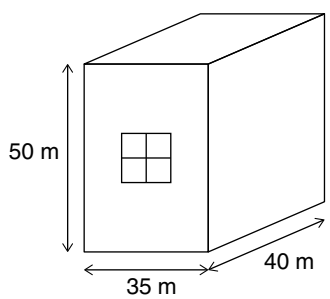


Fig f

Find the maximum number of windows, each of size 2 m × 3 m, that can be installed on the walls of the building if the equivalent temperature difference between the interior and the exterior of the building is 10 °C.

Given: U-value of the concrete of the building = 2.0 W m^{-2} K^{-1}

U-value of the glass of the windows = 5.7 W m^{-2} K^{-1}

A 960

B 598

C 160 (48.5%)

D 120

Conventional questions

15 3.2 A power station generates and transmits electrical energy to an electric vehicle via a charging point. For 100 J of energy in the fuel, about 35 J of energy reaches the vehicle after generation and transmission. It is then consumed by the vehicle and 28 J of energy becomes the kinetic energy of the vehicle.

(a) Estimate the end-use energy efficiency of the electric vehicle. (2 marks)

(b) Estimate the overall energy efficiency in running the electric vehicle. (2 marks)

***16** 3.1 Heat is conducted through an iron rod (Fig g). The length and the cross-sectional area of the rod are 1.5 m and 6.5 cm^2 respectively. The amount of heat transferred from the hot end to the cold end in 5 minutes is 10 kJ. The thermal conductivity of iron is 80 $W\ m^{-1}\ K^{-1}$ and the temperatures at the two ends of the rod remain constant.

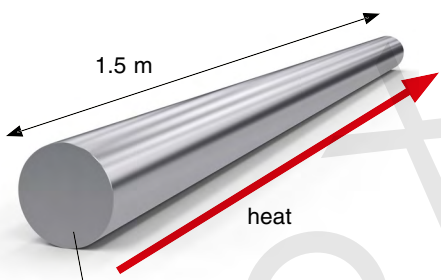


Fig g

(a) If the cold end of the rod is at 25 °C, what is the temperature at the hot end? (2 marks)

(b) A copper rod is of the same size as the iron rod. The temperatures at the two ends of the copper rod are the same as those of the iron rod. The thermal conductivity of copper is about 5 times that of iron. Estimate the amount of heat transferred through the copper rod in 5 minutes. (2 marks)

***17** 3.1 The thermal conductivity of concrete is 1.1 $W\ m^{-1}\ K^{-1}$. Sam wants to build a wall of U-value 0.5 $W\ m^{-2}\ K^{-1}$.

(a) If the wall is made of only concrete, how thick should the wall be? (2 marks)

(b) If the wall is made of only concrete, it will be too thick. Suggest and explain a method to build a thinner wall with the same U-value. (2 marks)

***18** 3.1 A single-glazed window and a double-glazed window are placed on the top of a bath of hot water at 70 °C. The lower surface of each window is in full contact with the water (Fig h). A temperature sensor is used to measure the temperature of the upper surface of each window. The experiment results are shown in Figure i.

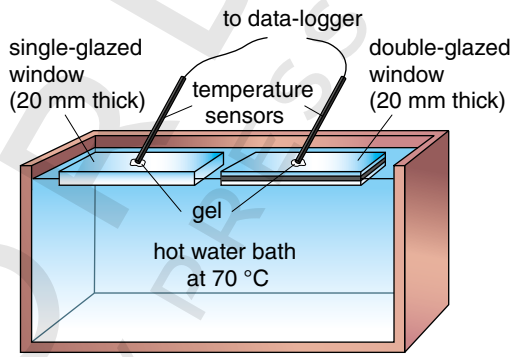


Fig h

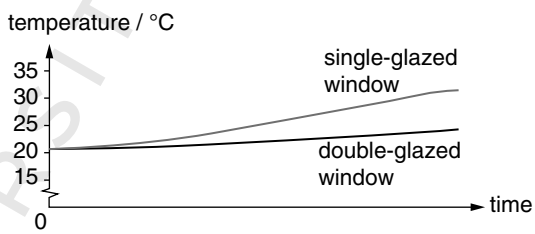


Fig i

(a) According to the experiment results, which window has a lower U-value? Explain. (2 marks)

(b) Explain why the window in (a) has a lower U-value. (2 marks)

(c) Some gel is applied between the tip of the temperature sensor and the upper surface of each window. Explain the function of the gel. (1 mark)

***19** 3.1 The table below shows the data for different parts of a building (Table a).

	Walls	Roof	Windows
Average rate of heat transfer per unit area / $W\ m^{-2}$	3	2	115
Total area / m^2	5000	600	1200

Table a

(a) Estimate the OTTV of the building. (2 marks)

(b) Suggest a way to reduce the heat transfer through

(i) the walls;

(ii) the windows;

(iii) the roof. (3 marks)

★20 3.2 Both hybrid vehicles and electric vehicles are equipped with regenerative braking systems.

(a) What is the function of a regenerative braking system? (1 mark)

(b) A car applies the regenerative braking system on a flat road. Describe the energy conversion involved. (2 marks)

(c) Another car applies the regenerative braking system when it moves down a slope at a constant speed (Fig j).

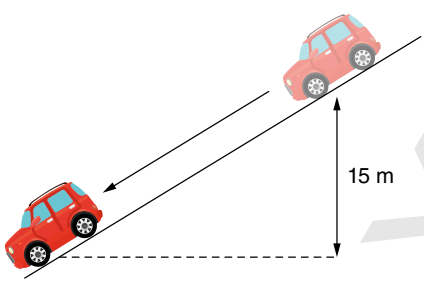


Fig j

The overall energy efficiency of the regenerative braking system is 32%. The mass of the car is 1800 kg. How much energy can be stored if the car moves by a vertical distance of 15 m? (2 marks)

(d) State two advantages of a hybrid vehicle over an electric vehicle. (2 marks)

★21 3.1 In countries with long periods of cold weather like Britain, solar control window films can be employed to reduce heat transfer from the interior to the exterior of the building.

(a) Give an advantage of solar control window films over tinted glasses. (1 mark)

(b) Should the films be installed on windows that receive more sunlight or less sunlight? Explain briefly. (2 marks)

(c) The installation of double-glazed windows is more important in Britain than in Hong Kong. Why? (2 marks)

★22 3.1

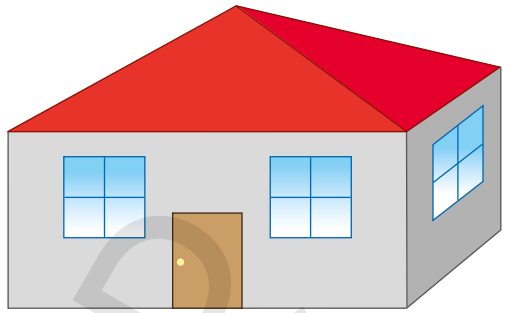


Fig k

Victor wants to estimate the rate of heat loss by conduction from his house on a winter day when the indoor temperature is 23 °C and the outdoor temperature is 3 °C (Fig k). He finds the data for the U-values of some common building components as in Table b.

Building component	U-values / W m^{-2} K^{-1}
Single-glazed window	5.5
Roof (no insulation)	2.0
Roof (with insulation)	0.5
Carpeted floor	0.8
Wall (no insulation)	1.7
Wall (with insulation)	0.6
Door (no insulation)	2.0
Door (with insulation)	1.0

Table b

(a) Table c shows the components of Victor's house. Complete Table c and find the total rate of heat loss by conduction on that day. (3 marks)

Surface	Total area / m^2	Rate of heat loss / W
6 single-glazed windows	12	
4 walls (no insulation)	75	
Roof (no insulation)		2800
Carpeted floors		960
Door (no insulation)	2	

Table c

(b) Electric heaters are used to maintain indoor temperature. Victor estimates that every 1000 W of heat loss costs $2.5 per day for heating. Installing insulation in the walls, the roof and the door costs $15 000. After how many days could the heating cost saved compensate the cost of installing the insulation? Assume the temperature difference is always maintained at 20 °C. (2 marks)

★**23** 3.1 According to the regulations in Hong Kong, the OTTV for a building envelope should not exceed 21 W m^{-2}.

(a) What is the meaning of the OTTV of a building envelope? (1 mark)

(b) Figure l shows the design of a cylindrical building. The height of the building is 63 m. The walls and the roof are made of concrete. The windows are distributed as shown. The U-value of the concrete part is 4.1 W m^{-2} K^{-1}. The rate of heat transfer per unit area through the windows is 140 W m^{-2}.

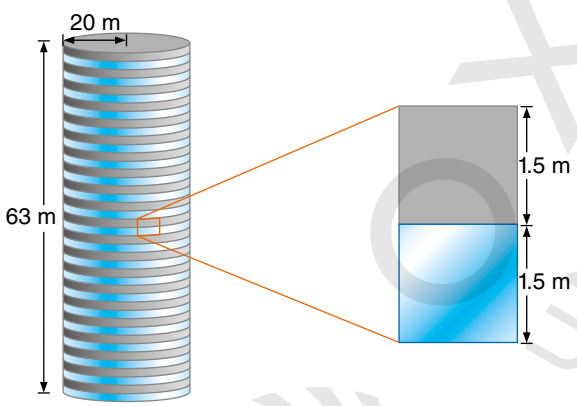


Fig l

(i) Does the OTTV of this building comply with the regulations in Hong Kong? Show your calculation. Take the equivalent temperature difference to be 6 K. (4 marks)

(ii) Suggest two methods to lower the OTTV of the building. (2 marks)

★**24** Misc

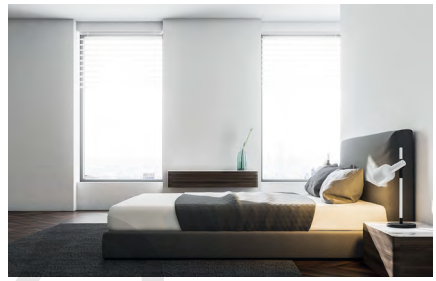


Fig m

A bedroom in a flat has only one wall facing outside. The wall has two windows (Fig m). The area of the wall (excluding the windows) is 10 m^2 and the total area of the windows is 6 m^2. An air conditioner is turned on and the room temperature is maintained at 25 °C. The temperature of the outside surface of the wall is 33 °C. The U-value of the wall is 1.2 W m^{-2} K^{-1}. The rate of heat transfer per unit area through the windows is 157 W m^{-2}. The rate of heat generated by indoor equipment is 500 W.

(a) Estimate the total rate of heat exchange between the inside and outside of the bedroom. Assume the heat transfer through the other walls, the ceiling and the floor is negligible. (4 marks)

(b) Estimate the cooling capacity of the air conditioner. (2 marks)

(c) Suggest three methods to reduce the energy consumption of the air conditioner. (3 marks)

★**25** 3.2 Figure n shows an electric vehicle used in a resort.



Fig n

The vehicle travels at 130 km h^{-1} when its power consumption is maximum, which is 47 kW. The maximum amount of energy stored in the battery of the vehicle is 16 kW h.

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(a) For how many hours can the vehicles travel with the maximum power output after the battery is fully charged? (2 marks)

(b) Estimate the range of the electric vehicle. (2 marks)

(c) Suggest two properties of an electric vehicle that makes it energy efficient. (2 marks)

(d) Jason says that using electric vehicles also contributes to the emission of greenhouse gases. Do you agree with him? Explain your point of view. (2 marks)

★★ **26** Misc

Figure o shows a Korean BBQ restaurant. The door and the windows are made of the same kind of glass and their total area is 60 m^2. The temperature of the exterior surface of the building envelope is 32 °C. An air conditioner is turned on so that the interior of the restaurant is maintained at 25 °C.

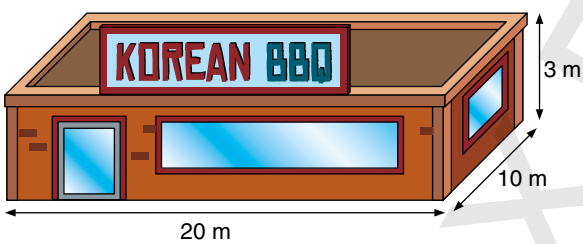


Fig o

(a) The walls and the roof are 20 cm thick and they have the same U-value of 1.2 W m^{-2} K^{-1}. The heat transfer per unit area through the windows and the door is 100 W m^{-2}. Estimate the total rate of heat transfer through the building envelope. (3 marks)

(b) There are 10 electric grills in the restaurant. Each grill releases heat inside the restaurant at a rate of 800 W. Assume there is no other heat source in the restaurant.

(i) How is the OTTV of the building affected by the grills? (1 mark)

(ii) If all the grills are turned on, estimate the cooling capacity of the air conditioner. (2 marks)

Refer p.97

27 3.1 **HKDSE 2017 Paper 2 Q.3(b)**

A refrigerated truck is used for transporting frozen goods (Fig p). A refrigerator is installed in the refrigerated compartment.



Fig p

Figure q shows the dimensions of the refrigerated compartment. The compartment is insulated using 0.08 m thick polystyrene. The thermal conductivity of polystyrene is 0.03 W m^{-1} K^{-1}.

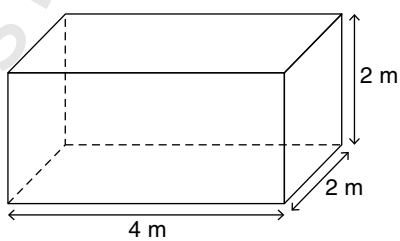


Fig q

(a) If a temperature difference of 50 °C is maintained between the exterior and the interior surfaces, estimate the minimum cooling capacity required for the refrigerator. (Hint: consider all the surfaces of the compartment.) (3 marks)

(b) On a sunny afternoon, the **air temperature** is 35 °C. By using the refrigerator with cooling capacity calculated in (a), briefly explain why the temperature inside the compartment **cannot** be maintained at −15 °C. (2 marks)

Q27 Exam report: Quite a number of them did not realize that the calculation involving thermal conductivity only dealt with heat transfer by conduction and thus failed to answer part (b) in which radiation had a part to play.

28 3.2 **HKDSE 2021 Paper 2 Q.3**

Some information of electric vehicles *A* and *B* is tabulated below (Table d):

Electric vehicle	Battery capacity / kW h	Maximum driving range / km	Mass / kg
A	95	326	2500
B	66	414	1620

Table d

(a) Although the battery capacity of *A* is higher, its maximum driving range is shorter than that of *B*. State a possible reason and explain why. (1 mark)

(b) (i) Suppose a charging voltage of 220 V is provided, estimate the minimum charging current required for charging the battery of vehicle *A* from completely discharged to fully charged in 12 hours. (2 marks)

(ii) Explain why in practice the charging current required is larger than that found in (b)(i). (1 mark)

The table below shows more information about the electric vehicles (Table e):

Electric vehicle	Time required to accelerate from 0 to 100 km h^{-1} / s	Peak power / kW
A	5.5	300
B	6.5	150

Table e

(c) Referring to all the information given, estimate

(i) the energy efficiency of vehicle *A*. You may assume that the vehicle is operating at its peak power. (2 marks)

(ii) the average output power from the battery of vehicle *B* if its average speed is 70 km h^{-1} in the maximum driving range test. (2 marks)

(d) Discuss in which of the driving modes below the regenerative braking system has the highest effectiveness: (2 marks)

Mode 1	driving at a few km per hour in often stop-and-go traffic conditions
Mode 2	driving in a city with smooth traffic regulated by traffic lights
Mode 3	driving on a highway

Q28 Exam report: In (a), very few candidates were able to point out that a more massive vehicle required a larger force/power for acceleration/deceleration. A common misconception was that a larger force/power was for maintaining the speed of a more massive vehicle.

Part (b) was well answered. A few candidates mistook 220 V as the peak voltage while weaker ones treated kW h as a unit of power.

Candidates did poorly in (c). As candidates had to tackle the questions from first principles using relevant data from the information given, not many were able to answer (c)(i)(ii) correctly. Weaker ones mixed up input and output powers or made mistakes in unit conversions.

Candidates' performance in (d) was fair. It seems that they did not fully understand how a regenerative braking system works. Quite a number of them wrongly chose Mode 3 (driving on a highway) to be the situation in which regenerative braking would be the most effective among the situations given.